

Dissecting infant leukemia developmental origins with a hemogenic gastruloid model

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eLife Assessment

This **valuable** study presents a mouse gastruloid system to generate successive waves of hematopoietic progenitors that in vivo would emerge during embryonic development. Although this newly revised manuscript has addressed some of the concerns raised during the first round of review, the study is still considered **incomplete**, as the claims are only partially supported. In particular, the claim of definitive wave hematopoietic progenitors being produced in the gastruloids, and their engraftment after transplantation, would benefit from further validation.

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Abstract

Current in vitro models of developmental blood formation lack spatio-temporal accuracy and weakly replicate successive waves of hematopoiesis. Herein, we describe a mouse embryonic stem cell (SC)-derived 3D hemogenic gastruloid (haemGx) that captures multi-wave blood formation, progenitor specification from hemogenic endothelium (HE), and generates hematopoietic progenitors capable of short-term engraftment of immunodeficient mice upon maturation in an in vivo niche. We took advantage of the haemGx model to interrogate the origins of infant acute myeloid leukemia (infAML). We focused on MNX1-driven leukemia, representing the commonest genetic abnormality unique to the infant group. Enforced MNX1 expression in haemGx promotes the expansion and in vitro transformation of yolk sac-like

erythroid-myeloid progenitors at the HE-to-hematopoietic transition to faithfully recapitulate patient transcriptional signatures. By combining phenotypic, functional and transcriptional profiling, including at the single-cell level, we establish the haemGx as a useful new model for the study of normal and leukemic embryonic hematopoiesis.

Introduction

Blood formation in the embryo is a multi-stage process that develops across multiple cellular niches, reflecting a complex hierarchy of tissue interactions. It involves sequential production of distinct cell types in so-called hematopoietic waves, separated in time and space (Lacaud and Kouskoff, 2017 [↗](#); Costa, Kouskoff and Lacaud, 2012 [↗](#); Medvinsky, Rybtsov and Taoudi, 2011 [↗](#); Dzierzak and Bigas, 2018 [↗](#)). The first wave of hematopoiesis produces unipotent red blood cell and macrophage precursors in the yolk sac (YS) from an angioblast precursor which can also form endothelium; it occurs at mouse embryo day (E)7-E7.5 (Fujimoto et al., 2001 [↗](#); McGrath and Palis, 2005 [↗](#); Moore and Metcalf, 1970 [↗](#)). The subsequent wave, known as definitive, produces: first, oligopotent erythro-myeloid progenitors (EMPs) in the YS (E8-E8.5); and later myelo-lymphoid progenitors (MLPs – E9.5-E10), multipotent progenitors (MPPs – E10-E11.5), and hematopoietic stem cells (HSCs – E10.5-E11.5), in the aorta-gonad-mesonephros (AGM) region of the embryo proper (Ivanovs et al., 2011 [↗](#); McGrath et al., 2015 [↗](#); Palis et al., 1999 [↗](#); Medvinsky and Dzierzak, 1996 [↗](#); Medvinsky, Rybtsov and Taoudi, 2011 [↗](#)). Definitive blood waves are characteristically specified from a specialised hemogenic endothelium (HE) through a process of endothelial-to-hematopoietic transition (EHT) (Zovein et al., 2008 [↗](#); Li, W. et al., 2005 [↗](#)). HSC specification from HE is the last event in hematopoietic development and occurs in characteristic cell clusters found on the ventral wall of the dorsal aorta (Ivanovs et al., 2011 [↗](#); Boisset et al., 2010 [↗](#)). All blood cell types enter circulation from their original locations and migrate to the fetal liver (FL) (Ciriza et al., 2013 [↗](#); Ghiaur et al., 2008 [↗](#)). Most embryonic blood progenitors are transient and sustain blood production exclusively during embryonic development, through differentiation. In contrast, HSCs make little or no contribution to embryonic and fetal blood: instead, they proliferate (i.e. self-renew) to expand their numbers. At least a subset of HSC migrates to the bone marrow (BM) niche at the end of gestation and balances self-renewal and differentiation to maintain blood production throughout the entire post-natal life (Bowie et al., 2006 [↗](#); Kikuchi and Kondo, 2006 [↗](#); Ganuza et al., 2022 [↗](#); Yokomizo et al., 2022 [↗](#)).

Blood progenitors and HSCs can be targeted by genetic alterations that confer clonal advantage and/or drive initiation and development of leukemia. Leukemia is the most common malignancy and the most common cause of cancer-related death in the paediatric age group (Steliarova-Foucher et al., 2017 [↗](#); Dong et al., 2020 [↗](#)). Several paediatric leukemias originate *in utero*, and a subset of these are driven by mutations exclusive to the paediatric group (Masetti et al., 2015 [↗](#); Cazzola et al., 2021 [↗](#)). Such paediatric-exclusive mutations target blood cell types restricted to development or depend on signals that are specific to embryonic hematopoietic niches, effectively defining distinct biological entities to adult malignancies (Cazzola et al., 2021 [↗](#); Bolouri et al., 2018 [↗](#); Chaudhury et al., 2018 [↗](#)). Dissection of these forms of developmental leukemia is confounded by the inability to robustly identify and isolate their embryonic cells of origin, and/or to fully recapitulate the disease in post-natal cells, which leads to failure in generating relevant targeted therapies (Alexander and Mullighan, 2021 [↗](#)). One such example is t(7;12)(q36;p13) acute myeloid leukemia (AML) driven by rearrangement and overexpression of the *MXI1* gene locus: *MXI1*-rearranged (*MXI1-r*) AML is restricted to infants under the age of 2 and carries a high risk of relapse and poor survival (Ragusa et al., 2023 [↗](#)). *MXI1* is a developmental gene required for specification of motor neurons and pancreatic acinar cells (Thaler et al., 1999 [↗](#); Harrison et al., 1999 [↗](#)). *MXI1* is not known to participate in embryonic hematopoiesis, however it has been discordantly described to be expressed in some subsets of adult hematopoietic cells (Deguchi and Kehrl, 1991 [↗](#); Nagel et al., 2005 [↗](#); Wildenhain et al., 2012 [↗](#); Taketani et al., 2008 [↗](#)). It is ectopically expressed in AML cells carrying one of a variable set of t(7;12) translocations which

place the normally silent *MNX1* gene under the control of *ETV6* regulatory elements (Weichenhan et al., 2023 [↗](#); Bousquets-Muñoz et al., 2024 [↗](#); Ragusa et al., 2023 [↗](#); Ballabio et al., 2009 [↗](#)). *ETV6* is required for hematopoietic cell specification from HE via VEGF signalling (Ciau-Uitz et al., 2010 [↗](#)), potentially positioning *MNX1* within a critical embryonic hemogenic regulatory network.

Faithful *in vitro* recapitulation of developmental blood cell types and their embryonic niches has the potential to dissect the cellular and molecular mechanisms presiding at initiation of developmental leukemia, reveal prognostic biomarkers, and indicate therapeutic vulnerabilities which are otherwise challenging to test in the embryo. Despite widespread use of embryonic stem (ES) cell and induced pluripotent stem (iPS) cell-based models of blood specification, a system which recapitulates the spatial and temporal complexity of embryonic blood formation within the respective time-dependent niches is still lacking (Dijkhuis et al., 2023 [↗](#)). In the last few years, gastruloid models have emerged as robust avatars of early development, self-organising processes of symmetry breaking, elongation, multi-axis formation, somitogenesis and early organogenesis, with remarkable similarity to embryo processes in space and time (van den Brink, Susanne C and van Oudenaarden, 2021 [↗](#); Beccari et al., 2018 [↗](#); van den Brink, Susanne C et al., 2020 [↗](#)). Gastruloids are grown individually in a multi-well format that facilitates tracking and screening of developmental processes. Relevant to blood formation, a variant gastruloid culture system, which successfully organises a heart primordium (Rossi et al., 2021 [↗](#)), was also able to recapitulate YS blood formation with erythroid and myeloid cellularity (Rossi et al., 2021 [↗](#); Rossi et al., 2022 [↗](#)).

Herein, we describe a new protocol of gastruloid formation, i.e. hemogenic gastruloids (haemGx), that captures putative YS-like and AGM-like waves of definitive blood progenitor specification from HE. The earlier wave of putative YS-like hematopoiesis includes EMP-like cells which further mature in the gastruloid culture. HaemGx AGM-like hematopoiesis captures the emergence of intra-aortic cluster-like cells, including cells capable of short-term multi-lineage hematopoietic engraftment of the spleen and BM of immunodeficient mice. We harnessed the embryonic hematopoietic context provided by haemGx to attest its utility in modelling leukemias with developmental origins, specifically by recapitulating *MNX1-r* infant leukemia by proxy of *MNX1* overexpression. *MNX1*-overexpressing haemGx showed expansion of cells at the HE-to-EMP transition, which became susceptible to transformation *in vitro* and recapitulated *MNX1-r* AML patient signatures, placing *MNX1*'s leukemogenic effects at a very early stage of developmental blood formation.

Results

HaemGxs capture time-dependent emergence of endothelium and hematopoietic cells

With the aim of recapitulating embryonic hematopoiesis with spatial and temporal accuracy, we modified existing gastruloid formation protocols to (1) promote the specification of hemogenic mesoderm derivatives, and (2) extend developmental time beyond the E8.0-equivalent of early developmental gastruloids (Van den Brink, Susanne C et al., 2014 [↗](#)) and the E9.0 timepoint of cardiogenic gastruloids (Rossi et al., 2022 [↗](#)). We established a 216h haemGx protocol (Fig. 1A [↗](#)) using a *Flk1-GFP* mouse ES cell line (Jakobsson et al., 2010 [↗](#)), which allows the tracking of early hemato-endothelial specification by expression of the *Flk1* (i.e. *Kdr*) locus. The haemGx protocol includes an indispensable pulse of Activin A to induce hemato-endothelial programmes via TGF- β signalling (visible by proxy of GFP-tagged *Flk1* activation), in addition of the characteristic gastruloid patterning via WNT activation by CHI99021 supplementation at 48h (Fig. 1S1A-B). We promoted hemato-endothelial programs through addition of VEGF and FGF2 at 72h (Sroczynska et al., 2009 [↗](#)) (Fig. 1A [↗](#)). We observed extension of a polarised *Flk1-GFP*-expressing branched endothelial network (Fig. 1B [↗](#)). Flow cytometry analysis revealed the onset of VE-cadherin⁺ C-Kit⁺ cells suggestive of EHT, with a peak at 120-144h (Fig. 1C-D [↗](#)). This was followed by a surge in

CD41⁺ candidate hematopoietic cells at 144h (**Fig. 1C-D**). In our experience, observation of early polarised activation of the *Flk1* locus at 96h (**Fig. 1B**) was predictive of CD41⁺ detection 2 days later, suggesting that TGF- β activation in early patterning is required for later hematopoietic development, putatively by modifying mesoderm specification. We incorporated a 24-hour pulse of sonic hedgehog (Shh) between 144-168h to mimic the aortic patterning that precedes cluster formation (**Fig. 1A**) (Rybtsov et al., 2014). Continued exposure to VEGF after the Shh pulse resulted in specification of CD45⁺ hematopoietic cells progenitors (**Fig. 1C-E**), which were apparent after 168h. The proportion of CD45⁺ cells was enhanced by the addition of SCF, FLT3L and TPO (**Fig. 1S1C-D**), a combination of cytokines routinely used for HSC maintenance; conversely, it did not require continued addition of FGF2 after 168h (**Fig. 1S1E**), which was omitted. CD45⁺ cells, some of which co-expressed C-Kit, were present in a small number of ‘cluster’-like structures adjacent to Flk1-GFP⁺ Kit⁺ endothelium (**Fig. 1E**). The process is overall reminiscent of hematopoietic progenitor cell budding from the aortic HE, suggesting recapitulation of hemogenic tissue architecture. Time-dependent specification of C-Kit⁺, CD41⁺ and CD45⁺ cells in haemGx could be obtained with other mouse ES cell lines, including commonly used E14Tg2a (E14) cells (**Fig. 1S1F-H**), with successful specification of comparable frequencies of CD45⁺ progenitors (**Fig. 1S1I**), attesting to the reproducibility of the protocol across cell lines of different genetic backgrounds.

Overall, our extended haemGx model could capture time-dependent specification of CD41⁺ and CD45⁺ hematopoietic cells, with recapitulation of the cluster-like spatial arrangement of CD45⁺ cell emergence from HE, thus showing initial promise as a tool for the exploration of normal and perturbed developmental hematopoiesis.

HaemGx specify cells with phenotypic and functional hematopoietic progenitor characteristics

We confirmed the induction of endothelial patterning from the Flk1-GFP population by immunofluorescence staining showing the branched vascular-like network of Flk1 co-expressing CD31 (**Fig. 2A**). Endothelial identity of Flk1-GFP⁺ populations was also detectable by flow cytometry through the presence of endothelial markers CD31 and CD34 (**Fig. 2S1A-B**). Given the co-expression of CD31 and CD34 with hematopoietic progenitors emerging from HE, we inspected CD41⁺ and CD45⁺ cells for the presence of these markers. Indeed, we found that subsets of CD41⁺ and CD45⁺ hematopoietic cells co-express CD31 and CD34 (**Fig. 2B**, **Fig. 2S1C-D**), with increased relative representation of hemato-endothelial markers within CD41⁺ cells at 144h and CD45⁺ at 216h (**Fig. 2C**), consistent with two successive waves of hemogenic emergence from HE. CD45⁺ cells from 216h haemGx also co-expressed C-Kit, Flk1-GFP and VE-cadherin, a phenotype enriched in emergent hematopoietic stem and progenitor cells (HSPC) in the AGM (**Fig. 2S1E-F**). The phenotype is detected in 1.8% of CD45⁺ cells on average (**Fig. 2S1G**), corresponding to an estimated 8 cells per haemGx. In agreement, analysis of the haemGx progenitor content at 216h using multipotential CFC assays (**Fig. 2D-E**) revealed the presence of >50 CFC/haemGx, of which an average of 7 have mixed-lineage granulocytic-monocytic-erythroid (GEM) potential, numerically matching the estimated number of HSPC. Other progenitors were predominantly granulomonocytic (GM) and erythroid (E) progenitors (**Fig. 2D-E**). Analysis of the cellularity of dissociated haemGx at the 216h timepoint using Giemsa-Wright stained cytopins revealed differentiated cells of G, M, E and megakaryocytic lineages (**Fig. 2S2A**), putatively reflecting the progeny of EMP-like cells.

We observed temporal progression of expression of key markers by real-time quantitative PCR (qPCR), configuring a sustained hemato-endothelial program from 120h through 216h (*Dll4* and *Mecom*), and the generation of hematopoietic populations based on the time-dependent patterns of *Myb*, *Runx1*, *Hlf* and *Hoxa9* (**Fig. 2F**). We note that *Myb* peaks at 144h, coincident with the emergence of CD41⁺ cells, which we postulate to have YS-like EMP characteristics. On the other hand, *Runx1* has a bimodal pattern peaking at 144h, and later at 216h, suggesting the presence of

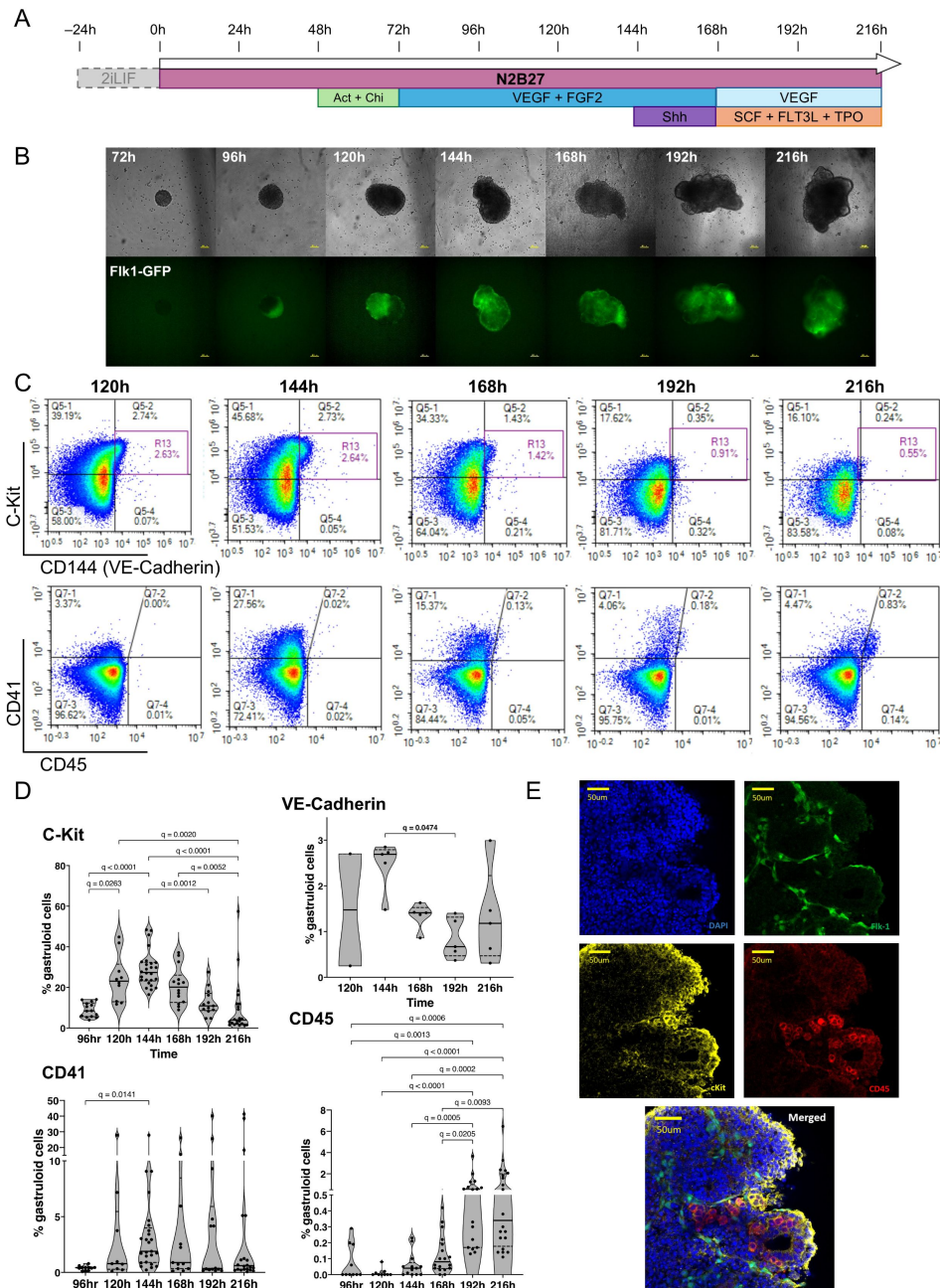


Figure 1

Hemogenic gastruloids (haemGx) produced from mES cells promote hemato-endothelial specification with spatio-temporally accurate ontogeny.

(A) Timeline of mES cells assembly and culture into gastruloids over a 216h period with the addition of specified factors for the promotion of hemato-endothelial specification. The 24-hour pre-treatment in 2i+LIF was omitted when mES cultures showed no signs of differentiation. **(B)** Imaging of haemGx over time from 72h to 216h at 10x magnification, showing the assembly and growth of the 3D structures and the polarization of the Flk-1-GFP marker from 96h; scale bar: 100µm. **(C)** Flow cytometry timecourse analysis of haemGx for the presence of CD144 (VE-cadherin), C-Kit, CD41, and CD45 markers; representative plots. **(D)** Quantification of analysis in (C). Violin plots with median and interquartile range of up to 26 replicates; Kruskal-Wallis CD144 ($p=0.0755$), C-Kit ($p<0.0001$), CD41 ($p<0.0342$), CD45 ($p<0.0001$). Shown are significant Dunn's multiple comparisons tests; adjusted q-value. **(E)** Immunostaining of whole individual haemGx at 216h showing the localized expression of Flk1-GFP (green), C-Kit (yellow), and CD45 (red) markers; nuclear staining is in DAPI (blue); scale bar: 100µm.

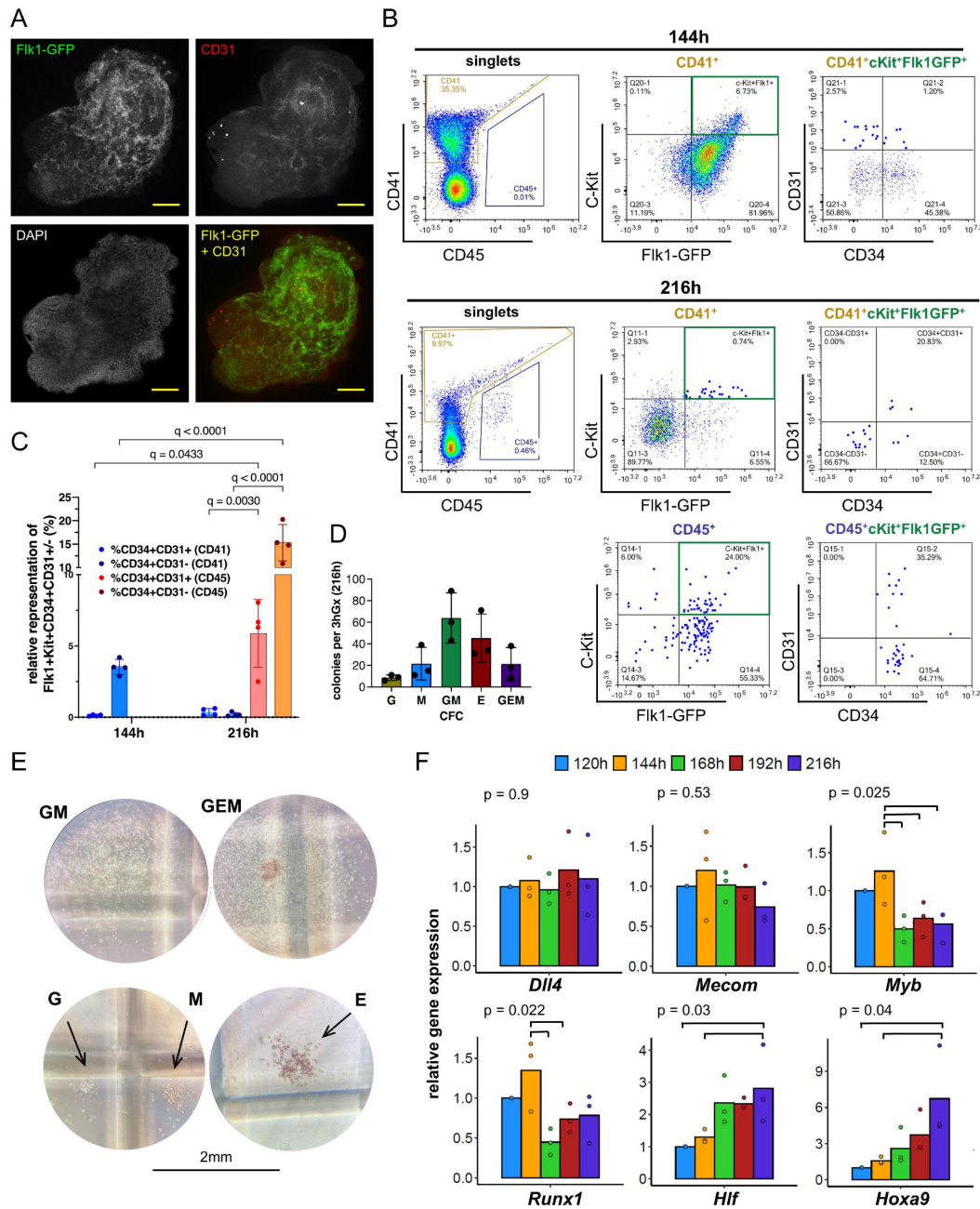


Figure 2

HaemGx produce hematopoietic output following the emergence of specialised endothelium.

(A) Immunostaining of whole individual haemGx at 144h showing the establishment of a vascular network by co-expression of CD31 and Fik1-GFP; nuclear staining is in DAPI (blue); scale bar: 100µm. **(B)** Flow cytometry analysis of haemGx cells at 144h and 216h stained for CD41, CD45, CD31, and CD34. **(C)** Quantification of flow cytometry analysis in (B) of CD41⁺ and CD45⁺ fractions co-expressing CD34 and CD31. Mixed effect analysis with a Šidák test for multiple comparisons. **(D)** Colony-forming cell (CFC) assay of 216h-haemGx in multipotential methylcellulose-based medium. GEM: granulocyte-erythroid-monocyte; GM: granulocyte-monocyte; M: monocyte; E: erythroid. Please note that the medium does not include TPO and does not assess the presence of megakaryocytic progenitors. CFC frequency in 3 haemGx, n=3, mean±SD. **(E)** Representative photographs of colonies quantified in C; scale bar=2mm. **(F)** Real-time quantitative (qPCR) analysis of expression of hematopoietic genes by timepoint. Relative gene expression fold change calculated by normalization to Ppia. Bars represent mean of 3 replicates and show individual data points; p values by ANOVA across datapoints; post-hoc statistical significance between specific variables by Tukey's test and shown by brackets.

two distinct hematopoietic waves. The later timepoint of 216h has significantly higher expression of both *Hoxa9* and *Hlf* raising the possibility that the second wave indeed corresponds to the specification of AGM-like HSPC. In order to further test the capture of YS-like and AGM-like hematopoietic waves, given the lack of markers that specifically associate with AGM hematopoiesis, we took advantage of the distinct regulatory requirements of polycomb-associated EZH2, which is necessary for EMP differentiation in the YS, but not for AGM-derived hematopoiesis or primitive erythroid progenitors (Neo et al., 2018 [DOI](#); Neo et al., 2021 [DOI](#)). Upon EZH2 inhibition via treatment with GSK216 from 120h onwards, we observed a reduction of CD41⁺ cells selectively at 144h, with no change in CD45⁺ cells at the later timepoint of 216h (Fig. 2S2B-C). This suggests that the cells present at the later timepoint may not be the progeny of the former, thus supporting recapitulation of 2 independent hematopoietic waves, the latter putatively AGM. Additionally, the vulnerability of 144h CD41⁺ to EZH2 inhibition suggests that they include YS-like EMP and do not predominantly reflect EZH2-independent primitive YS hematopoiesis.

HaemGx progenitors capture time-dependent signatures of YS-like and AGM-like hematopoiesis

To better characterize the extent and progression of developmental hematopoiesis in haemGx, we performed single-cell RNA-sequencing (scRNA-seq) time-course analysis of haemGx cell specification. We sorted cells from 2 independent haemGx cultures at 120, 144, 168, 192 and 216h, and profiled a total of 846 cells using the Smart-Seq2 protocol (Picelli et al., 2014 [DOI](#)) (Fig. 3S1A). We analyzed unfractionated live single haemGx cells at 120-192h and enriched the dataset with sorted CD41⁺ cells at their peak time of 144h, and sorted CD45⁺ cells at 192 and 216h. At timepoints of significant enrichment in CD41⁺ (144h) and CD45⁺ cells (192h) (Fig. 1D [DOI](#)), we also sorted C-Kit⁺ cells. Library preparation and sequencing generated an average of 120000 reads/cell, which were mapped to an average of 4000 genes/cell, with minimal signs of cell stress / death as measured by mitochondrial DNA fraction (Fig. 3S1B). Read and gene counts were similar between biological replicates, cell types and at different timepoints (Fig. 3S1C). The exception was 120h unfractionated haemGx cells, which despite similar sequencing depth (average read counts), were mapped to twice the number of genes (Fig. 3S1C), possibly reflecting multi-gene program priming at the onset of hemogenic specification. We selected highly varying genes (HVG) before principal component analysis (PCA) dimensionality reduction and retained the most relevant dimensions. In the PCA reduced space, we constructed a KNN graph and used uniform manifold approximation and projection (UMAP) to visualize the data on 2 dimensions, looking for cell communities using Leiden clustering (Fig. 3S1D). We identified 12 cell clusters of which 2 (clusters 0 and 5) almost exactly mapped to C-Kit⁺ cells (also positive for Sca-1), 2 (clusters 1 and 8) contained CD45⁺ sorted cells, and cluster 4 uniquely captured the CD41⁺ cells observed at 144h (Fig. 3S1D). Although some unfractionated cells overlapped with the hemogenic cell clusters (Fig. 3S1D, middle panel), most occupied different transcriptional spaces, reflecting the relative frequency of hematopoietic cells and suggesting the presence of potential hemogenic niches.

To explore tissue and lineage affiliation of the different clusters, we performed differential gene expression against all other cells using Wilcoxon ranking test, and established cluster classifier gene lists (Supplemental File S1). Top classifier genes for clusters 0 and 5 had endothelial affiliation (Supplemental File S1-S2), and the cellular composition of the clusters broadly reflected their time signature (Fig. 3S1D): cluster 5 comprised 144h cells and cluster 0 included most C-Kit⁺ cells at 192h. Cluster 5 occupied the vicinity of 144h CD41⁺ cells (Fig. 3S1D), which carried an erythroid-biased signature (Supplemental File S2). On the other hand, 192/216h-CD45⁺ cells in cluster 8 expressed myeloid and lymphoid genes (Supplemental File S1-S2). The remaining clusters largely corresponded to time-restricted unfractionated cells (Fig. 3S1D, bottom panel), with clusters of cells present at later timepoints matching mesenchymal stromal cell and autonomic neuron signatures (Fig. 3S1E), potentially configuring incipient formation of niches relevant to production of HSC (Kapeni et al., 2022 [DOI](#); Fitch et al., 2012 [DOI](#)).

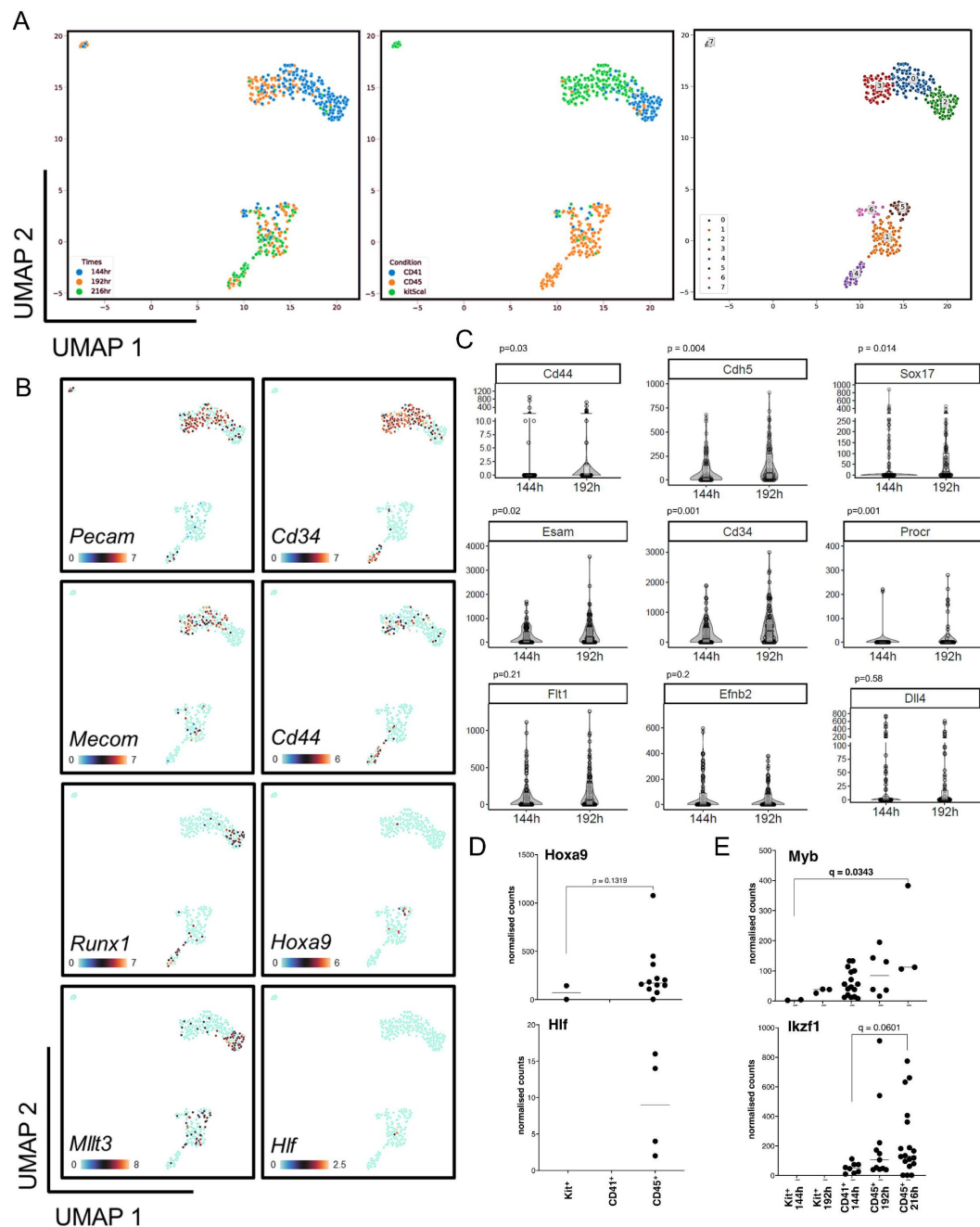


Figure 3

Time-resolved analysis of haemGx by scRNA-seq captures successive waves of hematopoietic specification.

(A) UMAP clustering of subsets of sequenced cells expressing CD41, C-Kit, and CD45 at 144h, 192h and 216h, annotated by timepoint (left panel), sorting condition (middle panel), and Leiden clustering (right panel). **(B)** UMAP panels highlighting the expression of specific hemato-endothelial markers in the transcriptional spaces defined in (A). Color scale indicates expression levels in counts. **(C)** Heatmaps of differentiation of an arterial endothelial programme in haemGx from scRNA-seq data of sorted on expression of C-Kit at 144h and 192h. Color scale indicates expression in counts. **(D)** Violin plots quantifying the expression of hemato-endothelial markers in C-Kit⁺ fraction from clusters in (A) comparing 144h and 192h timepoints. Wilcoxon test; significant p value<0.05. **(E)** Expression of definitive hematopoietic genes in in C-Kit⁺, CD41⁺ and CD45⁺ cells at 144h and 192/216h timepoints; expression levels as normalised counts per million reads. Mann-Whitney or Kruskal-Wallis with Dunn's multiple comparisons; significant p or q-value<0.05.

To specifically explore the extent to which haemGx capture putative YS and AGM-like hematopoiesis, we re-clustered cells sorted on hemato-endothelial markers CD41, C-Kit, and CD45 at the 144h, 192h and 216h timepoints (**Fig. 3A**). We mapped the expression pattern of endothelial (*Pecam* and *Cd34*), HE (*Mecom* and *Cd44*) and hematopoietic (*Runx1*, *Mllt3*, *Hoxa9*, and *Hlf*) markers across the time-dependent clusters (**Fig. 3B**). We observed a coincidence of *Pecam*, *Cd34*, and *Mecom* expression with clusters 0 and 3, which capture C-Kit⁺ cells at 144 and 192h, respectively. Some positive cells were also seen within CD41⁺ and CD45⁺-enriched clusters 2, and 1 and 4, respectively, supporting the presence of phenotypic progenitors, as suggested by flow cytometry analysis (see **Fig. 2**). CD41⁺ and CD45⁺-sorted clusters, on the other hand, show expression of *Runx1*, confirming their hematopoietic affiliation. *Cd44*, which is upregulated at the endothelial-to-hematopoietic transition (EHT) and in association with the AGM region (Oatley et al., 2020; Fadlullah et al., 2022), is detected predominantly in C-Kit⁺-enriched cluster 3 and CD45⁺-sorted clusters 1 and 4, corresponding to the later timepoints of 192 and 216h. This supports the notion that the later timepoints may capture AGM-like EHT, which is also suggested by significant increases in expression of *VE-cadherin* (*Cdh5*), *Sox17*, as well as *Cd34*, *Cd44* and *Procr* (Fadlullah et al., 2022; Jung et al., 2019; Oatley et al., 2020) in 192h, endothelial-enriched, C-Kit⁺ cells (**Fig. 3C-D**). Important in this context, these cells express arterial endothelial markers, including *Flt1*, *Efnb2* and *Dll4* steadily at both timepoints (**Fig. 3C** and 3S2B). *Hoxa9* expression also associates with arterial specification and definitive AGM-like hematopoiesis: albeit at low frequency, it is detected exclusively in C-Kit and CD45-sorted cells in the 192 and 216h-timepoints of haemGx differentiation (**Fig. 3B, E**). These cells also express *Mllt3* and, rarely, *Hlf*, compatible with incipient specification of an AGM-like definitive blood program (**Fig. 3B, E**). *Mllt3* is additionally expressed in CD41⁺-sorted cells of cluster 2, in line with its role in early erythroid and megakaryocytic lineage identity (Pina et al., 2008); *Myb* is also more frequently expressed in this population (**Fig. 3E**). In contrast, *Irf1*, which regulates lymphoid specification (Huang et al., 2019), gradually increases its expression, and indeed frequency, in 216h-CD45⁺ sorted cells. A handover from erythroid/megakaryocytic to myelo-lymphoid lineage affiliation is apparent between 144h and 216h haemGx blood cells (**Fig. 3S2C**). Of note, a small number of erythroid-affiliated cells at 216h express adult globins, suggesting a definitive origin (**Fig. 3S2C**).

Altogether, scRNA-seq analysis support the suggestion from phenotypic and *in vitro* functional data, that the haemGx model captures two waves of hematopoietic specification from HE-like cells and may configure pre-definitive and definitive embryonic blood formation.

Late-stage haemGx cells transcriptionally resemble mouse AGM progenitors and have short-term engraftment potential

To investigate the putative YS-like pre-definitive and AGM-like definitive nature of haemGx-generated hematopoietic progenitors, we conducted projections of haemGx sc-RNAseq data onto *bona fide* embryonic populations. We considered 3 distinct scRNA-seq datasets, which (1) capture arterial and haemogenic specification in the para-splanchnopleura (pSP) and AGM region between E8.0 and E11 (Hou et al. 2020) (**Fig. 4S1A**); (2) uniquely capture YS, AGM and FL progenitors and the AGM EHT (Zhu et al., 2020) (**Fig. 4S1B**); and (3) interrogate the mouse AGM at the point of HSC emergence from the dorsal aorta HE (Thambyrajah et al., 2024) (**Fig. 4S1C**). We extracted highly-varying genes from each of the 3 datasets to construct dimensionality reduction maps of the data (**Fig. 4S1A-C**), and conducted k-Nearest Neighbor (KNN) analysis to classify and regress our haemGx scRNA-seq data onto those maps considering 3 neighbors (**Fig. 4A**). HaemGx cells with correlation metric >0.7 were projected onto the respective embryonic datasets (Supplemental File S3). Projections onto Hou et al. (2020) positioned clusters 0 and 3, corresponding to endothelium, including arterial endothelium and HE-like cells, between days E8.5 and E10, putatively spanning YS-like and AGM-like blood formation. A subset of CD45⁺-sorted cells, as well as a small numbers of CD41⁺ cells from cluster 2 projected onto E10 HE and intra-aortic cluster (IAC) cells, suggesting the presence of definitive HSPC or their precursors in haemGx (**Fig. 4A**, Supplemental File S3). We sought to further check the transcriptional affiliation of haemGx with

intra-embryonic, definitive hematopoiesis, by considering the dataset by [Zhu et al. \(2020\)](#) [\(Fig. 4A\)](#), which uniquely captures different hematopoietic locations in the same study ([Fig. 4A](#)). We matched the identity of a large proportion of CD41⁺-sorted cells to YS EMP, with some of the same cluster 2 cells mapping to erythroid progenitors, and some later CD45⁺-sorted cells resembling myeloid progenitors ([Fig. 4A](#), Supplemental File S3). On the other hand, a subset of CD45⁺-sorted cells mapped to IAC and lymphoid-myeloid primed progenitors (LMPP) in the FL ([Fig. 4A](#)), compatible with the presence of AGM-like definitive HSPC in haemGx. Interestingly, some CD41⁺-sorted cells also map to IAC, and indeed to HE, suggesting that this earlier population may contain HSPC precursors, in addition to YS-like progenitors. IAC and HE-like cells were also found within C-Kit⁺ cells, mostly from the 192h timepoint, while the majority of C-Kit⁺ cells captured different endothelial identities, as expected from their transcriptional profiles. Projection onto [Thambyrajah et al. \(2024\)](#) [\(Fig. 4A\)](#), Supplemental File S3), conflates YS-like and AGM-like progenitors onto the candidate HSC-enriched embryo AGM cluster 2, highlighting the similarity between transcriptional programs at different sites of embryonic blood emergence. CD45⁺ cells are more distally-projected onto the same cluster, as shown for E11.5 cells in the [Thambyrajah et al. \(2024\)](#) study, some of which correspond to the candidate FL-LMPP in [Zhu et al. \(2020\)](#) (Supplemental File S3). HaemGx cell similarities to [Thambyrajah et al. \(2024\)](#) HSC-enriched cluster 2 also include HE-like and IAC-like cells (as per [Zhu et al. 2020](#) projections), mostly E9.5-like as per [Hou et al. \(2020\)](#), overall suggesting putative early specification of the AGM region within haemGx.

Late-stage haemGx cells can engraft hematopoietic tissues upon maturation

The presence of putative AGM-like hematopoietic cells within haemGx led us to investigate whether haemGx-derived cells had the ability to engraft the hematopoietic system of immunodeficient animals. In early experiments, we had transplanted dissociated haemGx cells obtained at 192 and 216h into irradiated immunocompetent C57Bl/6 recipients, sub-lethally irradiated immunodeficient NSG mice, or non-conditioned NSGW41 hosts, but failed to detect signs of engraftment of peripheral blood, spleen (Spl) or bone marrow (BM) at 12 days (including absence of spleen colony formation), at 4-6 weeks (short-term engraftment), or beyond 8-10 weeks (long-term engraftment) by either flow cytometry detection of CD45.2 or by genomic DNA (gDNA) PCR analysis of the *Flk1-GFP* locus.

Failure to engraft was perhaps not surprising given that transcriptional programs position haemGx no later than E9.5-E10, which precedes HSC specification in the AGM. Additionally, specification of HSC-supporting stroma within haemGx, namely mesenchymal cells ([Kapeni et al., 2022](#)) and elements of the sympathetic nervous system ([Fitch et al., 2012](#)), is only detectable at the 192h timepoint, suggesting that further 3D maturation *in vivo* may be required. We thus tested implantation of undissociated haemGx in the adrenal gland of *Nude* mice ([Fig. 4B](#)), a topography more commonly used to support tumor development, but that was recently shown to be able to support HSC development, possibly with contributions from the sympathetic adrenal medullary niche ([Schyrr et al., 2023](#)). In order to facilitate the analysis of engrafted animals, and given the coincidence of CD45 isoform between *Flk1-GFP* mouse ES cells and the *Nude* mice in which the experiments were performed, we engineered constitutive expression of BFP from the *Rosa26* locus in the *Flk1-GFP* mES cell line (*Rosa26-BFP::Flk1-GFP*) ([Fig. 4S2A](#)). We implanted 3 haemGx each unilaterally in the adrenal gland of unconditioned recipient mice ([Fig. 4B](#)). Control mice were injected with an equivalent volume of PBS. Animals were sacrificed at 4 or 8 weeks after implantation, with 4-5 experimental animals and 1 control collected at each timepoint. We checked Spl and BM BFP engraftment, by flow cytometry and by PCR analysis of DNA. Sensitivity of PCR analysis was determined by serial dilution of *Rosa26-BFP::Flk1-GFP* mouse ES cells in the human K562 cell line, with detection of as low as 10 pg of mouse input in a total of 100 ng gDNA ([Fig. 4S2B](#)), corresponding to approximately 3 cells (in 300,000). PCR analysis of BFP engraftment of the Spl and BM 4 weeks after adrenal implantation of 3 haemGx / animal detected

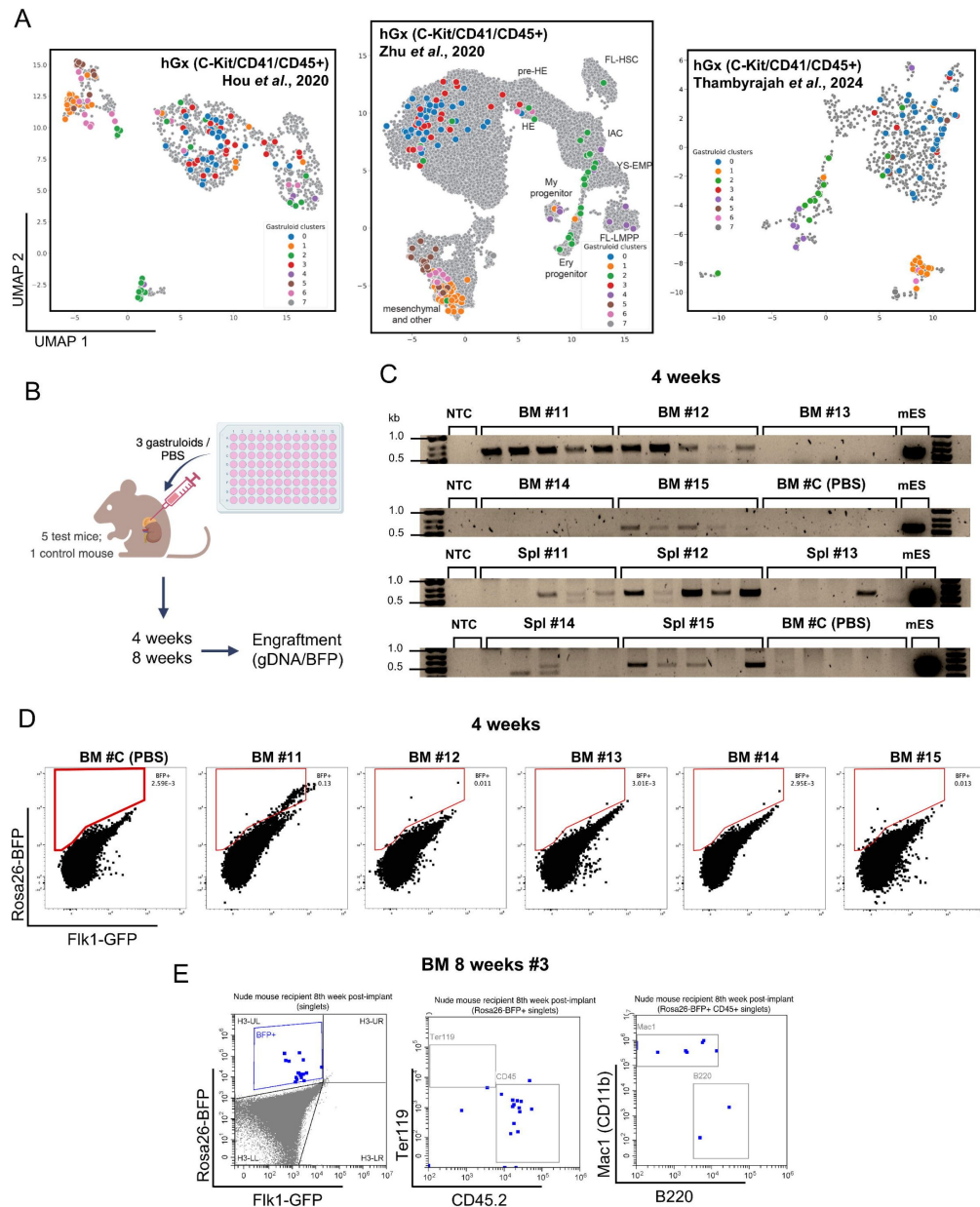


Figure 4

Late-stage haemGx contain hematopoietic output with transcriptional alignment to mouse embryonic populations and in vivo engraftment potential.

(A) Projection of clustered haemGx cells (CD41, C-Kit, and CD45) at 144h, 192h and 216h; **(Fig. 3A)** onto mouse single-cell RNA-seq datasets capturing: arterial and haemogenic specification in the para-splanchnopleura (pSP) and AGM region between E8.0 and E11 (Hou et al., 2020); YS, AGM and FL progenitors and the AGM EHT (Zhu et al., 2020); HSC emergence from the dorsal aorta HE (Thambyrajah et al., 2024). Gastruloid cell projection identified as per their cluster of origin (**Fig. 3A**). **(B)** Schematic representation of the experimental workflow for execution and analysis of haemGx implantation in the adrenal gland of immunodeficient mice. **(C)** PCR detection of haemGx genomic (g)DNA in the bone marrow (BM) and spleen (Spl) of immunodeficient mice 4 weeks after unilateral adrenal implantation of 3 gastruloids/gland. Analysis of 5 replicates of 100ng gDNA/recipient tissue; control animal was injected unilaterally with PBS in an adrenal gland in parallel with experimental implantation. Reaction positive control used 100ng of gDNA from *Rosa26-BFP::Flk1-GFP* mES cells (mES) used to generate haemGxs. NTC: no template control. **(D)** Flow cytometry plots of engraftment detection by BFP expression in bone marrow (BM) of recipient mice 4 weeks following implantation of 216h haemGx. **(E)** Flow cytometry plots of lineage affiliation of BFP⁺ engraftment in bone marrow (BM) of recipient mice 8 weeks following implantation of 216h haemGx.

BFP amplicon in both tissues in 3 of the 5 samples (#11, 12 and 15) (**Fig. 4C** and 4S2C). We sampled the gDNA of each recipient 5 times and detected the presence of the amplicon in 3-5 of the replicate samples of the positive animals; replicate sampling of the control was consistently negative (**Fig. 4C** and 4S2C). The same 3 animals showed BFP detection by flow cytometry above control, but no higher than 0.1% (**Fig. 4D**). PCR engraftment was nevertheless detectable in both erythroid Ter119⁺ and lympho-myeloid CD45⁺ fractions of the recipient animals showed PCR engraftment in both lineage fractions (Fig. 4S2D-E). Analysis of engraftment 8 weeks after implantation showed low-level detection (<0.1%) of BFP⁺ cells with positivity for CD11b⁺ myeloid and B220⁺ lymphoid compartments in BM of 1 recipient (**Fig. 4D** and 4S2F-G); a 4th recipient had to be sacrificed early due to infection.

Overall, implantation in the adrenal niche and/or preservation of the 3D architecture enabled low-level *in vivo* hematopoietic activity, although the mechanism remains unclear. In some recipients, hematopoietic engraftment was accompanied by *in situ* growth and multi-tissue differentiation of the implanted haemGx (Fig. 4S2H). However, adrenal enlargement was not observed in all animals with hematopoietic engraftment (e.g. 4 weeks' #11 had a normally sized adrenal, as did all animals analysed at 8 weeks), making it unlikely that hematopoietic activity results from teratoma formation by residual pluripotent cells. Late-stage haemGx (216h) may thus contain hematopoietic progenitors with some short-term engraftment potential but incomplete functional maturation, which are present at low frequency. Engraftment is erythro-myeloid at 4 weeks and lympho-myeloid at 8 weeks, reflecting different classes of progenitors, putatively of YS-like and AGM-like affiliation.

HaemGx resolve the early hemogenic ontogeny targeted by infant AML gene MNX1

The ability of haemGx to reconstitute putative YS and AGM-like HE and hemopoietic progenitors positions it as an attractive model to probe cellular origins of developmental leukemia. In particular, forms of AML unique to the first 2 years of life – infant (inf)AML – have been shown to transform fetal liver (FL) but not adult hematopoietic cells (Waraky et al., 2024; Mercher et al., 2009; Chen, W. et al., 2011; Ragusa et al., 2023), compatible with targeting of a transient embryonic cell. In the case of t(7;12) AML (herein MNX1-r) (Ragusa et al., 2023), in which the translocation results in transformation-driving ectopic expression of *MNX1* (Waraky et al., 2024), analysis of patient transcriptional signatures (Ragusa et al., 2022) for over-representation of cell atlas-defined affiliations, indicates enrichment in cardiomyogenic, endothelial, HSC/progenitor and mast cells (**Fig. 5A**). These signature enrichments are compatible with an early hemogenic cell derivative, capturable in the haemGx model. Other cell type-enrichments reflect MNX1 functions in pancreatic and neuronal development (Ragusa et al., 2022) (Fig. 5S1A).

We overexpressed *MNX1* (MNX1-OE) in *Flk1-GFP* mouse ES cells by lentiviral transduction (**Fig. 5B**) and analyzed progression of haemGx generation (**Fig. 5C**). We used human *MNX1* cDNA to distinguish from the endogenous gene, but the degree of homology is nevertheless high (84%), supporting functional equivalence. We confirmed that *MNX1* overexpression in haemGx is maintained to endpoint (Fig. 5S1B). MNX1-OE haemGx activated polarized *Flk1-GFP* expression and elongated with similar kinetics to empty-vector (EV) control (**Fig. 5C**), albeit with formation of larger haemGx (**Fig. 5D**) denoting increased cellularity (Fig. 5S1C). From 192h onwards, MNX1-OE haemGx had a higher frequency of spontaneously contractile structures (Fig. 5S1D), compatible with the cardiogenic cell association of patients' transcriptomes (**Fig. 5A**).

We interrogated the time-dependent hemogenic cell composition of haemGx by flow cytometry, quantifying C-Kit, CD41 and CD45 (**Fig. 5E-G**, 5S1E-F), through which we had captured endothelial/HE-like cells, YS-EMP-like, and later putative AGM-like progenitors, respectively. We

observed a small but significant expansion of the C-Kit⁺ compartment specifically at 144h (**Fig. 5E** [5S1E](#)). All cell markers were relatively unchanged at later timepoints, although absolute cell numbers were higher in MNX1-OE haemGx (**Fig. 5S1C**).

To better understand the consequences of MNX1-OE on hemogenic development, we performed RNA-sequencing (RNA-seq) of haemGx at 144h and 216h, capturing putative YS and AGM-like hemopoietic waves, respectively (**Fig. 5H** [5H](#)). We confirmed expression of the human *MNX1* transgene in the sequenced reads (**Fig. 5S2A**); in contrast, we could not detect endogenous mouse *Mnx1* in any of the samples (**Fig. 5S2A**), confirming that it does not normally play a role in hemogenic development. DEGs between MNX1-OE and EV (Supplemental File S4) configured distinct MNX1-driven programs at 144h and 216h, with >80% genes timepoint-specific (**Fig. 5H** [5H](#)). Both programs include up-regulation of hematopoietic-associated genes, capturing EHT and progenitor regulators (e.g. *Runx1*, and *Epor* and *Zfpm1*, respectively) at the earlier timepoint, and more differentiated effectors, e.g. *Hba-a1*, *Hbb-bh1* and *Il3ra* at 216h, compatible with selective targeting and/or expansion of YS EMP-like cells and their later differentiated progeny (**Fig. 5H** [5H](#)). An overview of transcriptional regulatory programs at both timepoints denotes a clearer hemogenic enrichment at 144h, with over-representation of GATA2, GATA1 and RUNX1 binding targets (**Fig. 5I** [5I](#)), putatively capturing an expansion of a HE-to-EMP-like transition, also represented in enriched GO categories related to VEGF signalling and epithelial-to-mesenchymal transition (EMT) (**Fig. 5J** [5J](#)). In contrast, no enrichment of hemogenic transcription factor (TF)-ChIP targets or related gene ontologies (GO) categories was observed in MNX1-OE haemGx at 216h (**Fig. 5I-J** [5I-J](#)), suggesting specific targeting of YS-like hemogenic programs. Instead, there was an enrichment in cardiogenic GO at 216h (**Fig. 5J** [5J](#)), reflecting the increase in contractile foci in MNX1-OE late haemGx.

We attempted to further deconvolve the cellular heterogeneity underlying bulk RNA-seq DEGs by interrogating the cluster signatures obtained from the single-cell analysis of haemGx development (**Fig. 5S2B-C**; refer to **Fig. 3S1D**). In support of an association with hemato-endothelial specification, MNX1-OE transcriptomes showed enrichment of haemGx HE cluster 5 and 0 signatures; the YS-EMP-like cluster 4 was also enriched (**Fig. 5S2B**). MNX1-OE DEGs at both timepoints also enriched for the relatively less differentiated cluster 10, which predominantly captures unfractionated haemGx cells at 120h, supporting an early ontogenic affiliation. Significantly, clusters 5 and 0 HE signatures were also enriched in *MNX1*-r infAML patient signatures (Ragusa et al., 2022 [5](#)) (**Fig. 5S2B-C**), putatively aligning MNX1-OE effects with HE and the hemogenic transition. In contrast, MNX1-OE DEGs specific to 216h showed enrichment in myeloid progenitor / MLP-enriched cluster 8 which captures 192/216h haemGx cells, but diverges from *MNX1*-r infAML signatures (**Fig. 5S2B-C**), instead approximating the distinct myelo-monocytic and/or mixed-lineage affiliation of *KMT2A*-rearranged AML, which is also common in older pediatric and adult patients.

Serial replating of MNX1-OE haemGx cells identifies a candidate C-Kit⁺ leukemia-propagating cell more susceptible to transformation in the YS period

We further investigated the potential of MNX1-OE haemGx to reflect *MNX1*-r AML biology by performing serial replating of CFC assays, a classical *in vitro* assay of leukemia transformation. We compared 144h and 216h haemGx to match functional and transcriptome data, and plated unfractionated haemGx, given the transient nature of C-Kit⁺ enrichment. Both 144h (**Fig. 6A-B** [6A-B](#)) and 216h (**Fig. 6C-D** [6C-D](#)) MNX1-OE haemGx replated for at least 5 platings, significantly different to control (EV) (**Fig. 6B,D** [6B,D](#)). EV-haemGx replating could not be sustained beyond plate 3 in most experiments started from 216h (**Fig. 6C** [6C](#)); 144h EV-haemGx exhibited some low-level replating until plate 5 (**Fig. 6A** [6A](#)), but the colonies were formed by few scattered cells clearly distinct from the well-demarcated colonies obtained from MNX1-OE haemGx. We analyzed early-stage (plate 1) and late-stage (plate 5) colonies by flow cytometry to identify the nature of the replating cells. Cells

selected through replating were C-Kit⁺ (**Fig. 6E**), matching the cell phenotype transiently enriched at 144h. MNX1-OE C-Kit⁺ cells co-expressed CD31 (**Fig. 6S1A-B**), underpinning the endothelial signatures of *MNX1-r* leukemia. Early replating of MNX1-OE haemGx also enriched for erythroid-affiliated Ter119⁺ cells, particularly when initiated from the 144h timepoint (**Fig. 6S1C**), but this was not sustained through replating. Myelo-monocytic-affiliated markers, including CD11b or CD45, were not present (**Fig. 6E**, **6S1C**).

We performed RNA-seq analysis of MNX1-OE haemGx CFC cells obtained through re-plating and integrated the data with the 144h and 216h haemGx timepoints to identify genes enriched through the process of transformation. Hs*MNX1* reads were increased in MNX1-OE CFC samples, supporting selective expansion of transduced cells (**Fig. 7A**). We performed hierarchical clustering of all DEGs in pairwise comparisons (**Fig. 7B**), and used Gene Set Enrichment Analysis (GSEA) of *MNX1-r* patient signatures up-regulated in comparison to other AML in the infant (0-2) age group, to interrogate the different clusters. Cluster k14 had significant enrichment of *MNX1-r* signatures against all other AML subtypes (**Fig. 7C**), with considerable overlap between leading-edge (i.e. significantly-enriched) genes (**Fig. 7S1A**). Importantly, k14 captures the subgroup of DEGs up-regulated upon CFC replating which are also up-regulated in MNX1-OE haemGx at 144h (**Fig. 7B**), matching the time-dependent enrichment of the putative MNX1-OE propagating population. Interrogation of cell-type-specific signatures within the k14 MNX1-*r* leading edge genes against the single-cell PanglaoDB atlas captured similarities to hemogenic and non-hemogenic cell types, including interneuron, and other neural signatures (**Fig. 7D**), which reflect MNX1 biological functions (Harrison et al., 1999; Thaler et al., 1999; Ragusa et al., 2022). The most highly significant signatures capture early developmental hemogenic components, such as ‘endothelial cells’ and ‘erythroid cell precursors’, reflecting the 144h, putative YS affiliation of MNX1-OE haemGx. The ‘mast cells’ affiliation of *MNX1-r* AML signatures within the haemGx k14 cluster is also likely to reflect a YS EMP-like affiliation (Chia et al., 2023), and is accordingly aligned with MNX1-OE haemGx isolated at 144h (**Fig. 7S1B**). Giemsa-Wright-stained cytopins of serially-replated CFC assays initiated with 144h MNX1-OE cells showed a shift from blast-like at plate 3 to precursor cell morphology with some differentiated mast cells at plate 5 (**Fig. 7E**, **S1C**). We also observed mast cells in 216h-initiated colonies (**Fig. 7E**), but these were differentiated at an earlier plate, suggesting that MNX1-OE-responsive C-Kit⁺ progenitors experience maturation between 144h and 216h.

Taken together, the data positioned the initiating cellular targets of *MNX1-r* AML at the YS endothelial-to-EMP transition. More broadly, the analysis supports the potential of haemGx to dissect the ontogeny of specific subtypes of hematological cancers initiated *in utero*.

Discussion

In this study, we developed and characterized a hemogenic gastruloid (haemGx) model from mouse ES cells, which recapitulates YS-like and AGM-like waves of definitive blood progenitor specification, and has incipient ability to engraft adult hematopoietic tissues short-term upon *in vivo* maturation. By exploring the haemGx model through enforced expression of the *MNX1* oncogene, which is a hallmark of infant AML harbouring the t(7;12) translocation, we identified the YS endothelial-to-hematopoietic transition as the transformation target of *MNX1* gene rearrangement, shedding light on to its exclusive association with the infant age group.

Our data positions the haemGx model as an advanced platform to probe physiological and pathological development of the hematopoietic system, that benefits from the time-dependent and topographic organization associated with gastruloid models. We constructed the haemGx model by introducing a pulse of Activin A and extending the protocol under hematopoiesis-promoting cytokines, to reach the equivalent of mouse embryonic day 10 (E10) and achieve sequential acquisition of YS-like and AGM-like blood production from HE, benchmarked by surface markers

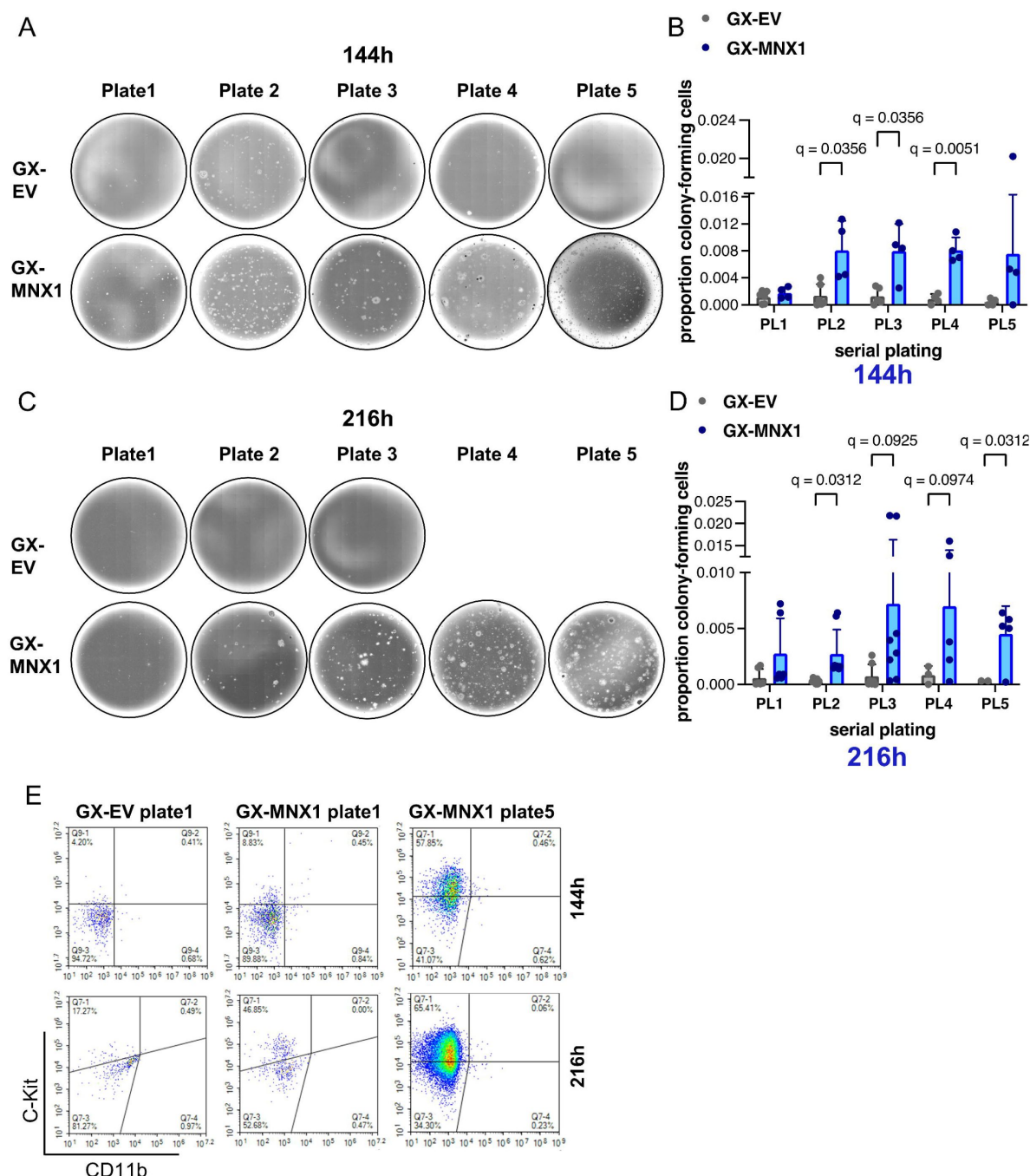


Figure 6

MNX1 selects C-Kit⁺ clonogenic cells and selectively transforms end-stage haemGx.

(A) Representative photographs of serial replating of colony-forming cell assays initiated from EV or MNX1 cells obtained at 144h of the haemGx protocol. (B) Quantification of colony-replating efficiency of EV and MNX1 144h-haemGx (GX) cells. Mean+SD of $n > 3$ replicates; Kruskal-Wallis with Dunn's multiple comparison testing at significant $q < 0.05$. (C) Representative photographs of serial replating of colony-forming cell assays initiated from EV or MNX1 cells obtained at 216h of the haemGx protocol. (D) Quantification of colony-replating efficiency of EV and MNX1 216h-haemGx cells. Mean+SD of $n = 5-8$ replicates; Kruskal-Wallis with Dunn's multiple comparison testing at significant $q < 0.05$. (E) Flow cytometry analysis of hemogenic progenitor C-Kit⁺ cells and myeloid-affiliated CD11b⁺ cells at first round of plating of 144h and 216h-haemGx initiated from EV and MNX1 cells; representative plots.

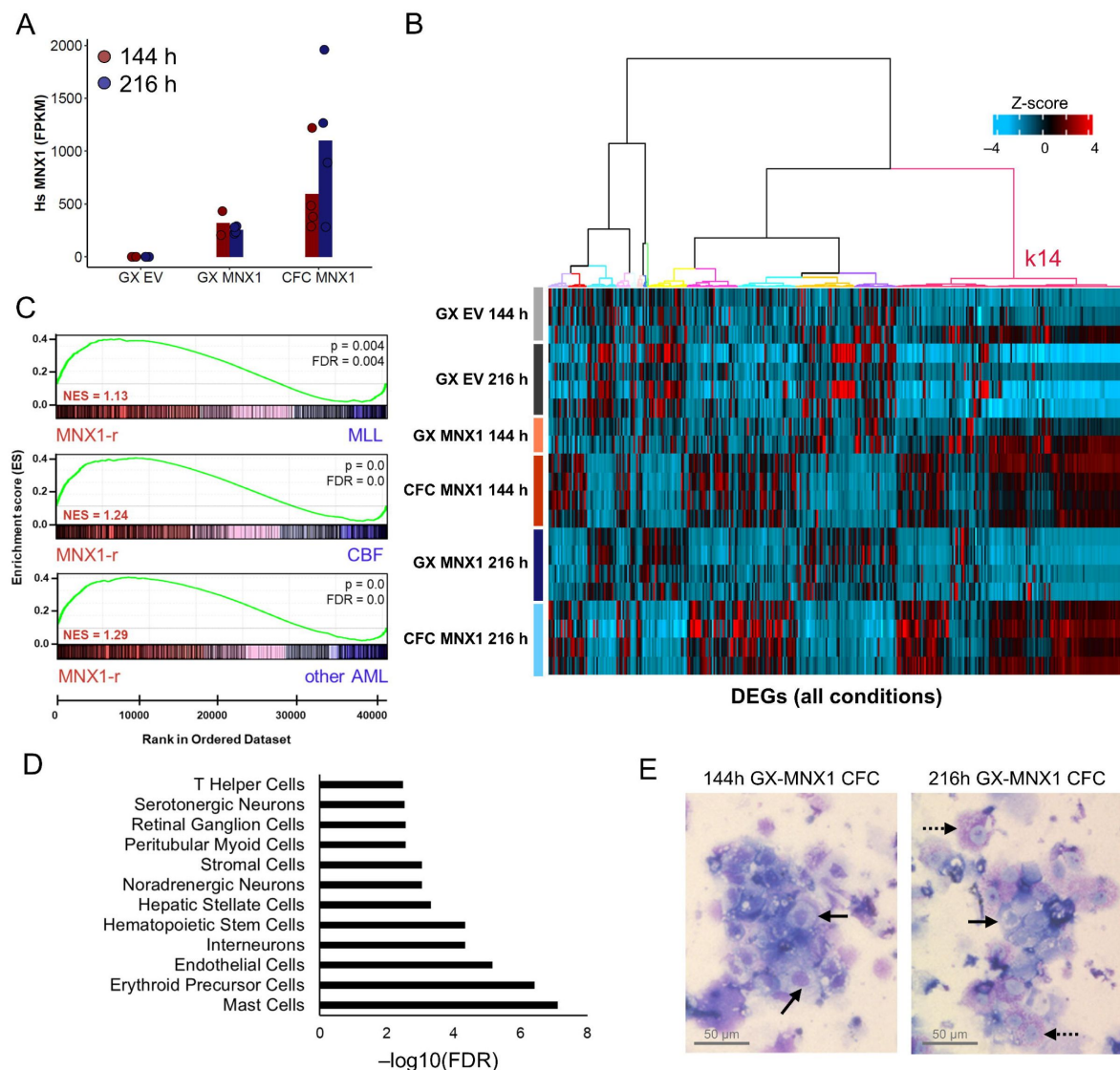


Figure 7

HaemGx with *MNX1* overexpression recapitulate *MNX1*-Acute Myeloid Leukemia patient signatures.

(A) Quantification of human *MNX1* transcripts in FPKM units from RNA-seq of GX-EV, GX-MNX1, and CFC at 144 hr and 216 hr. Bars represent mean of replicates. (B) Heatmap comparing the expression of all differentially expressed genes (DEGs) between all conditions (GX-MNX1 144 hr vs GX-EV 144 hr; GX-MNX1 216 hr vs GX-EV 216 hr; CFC MNX1 144 hr vs GX MNX1 144 hr; CFC MNX1 216 hr vs GX MNX1 216 hr) as Z score. Hierarchical clustering by Ward D method on Euclidean distances identifies 14 clusters (k). (C) GSEA plots for gene set extracted from k14 from (B) against MNX1-r patients RNA-seq counts vs MLL, core-binding factors (CBF), or other pediatric AML. (D) Cell type analysis of the intersect of leading edge genes LEGs (n=476) from Fig. S5D in MNX1-r patients (k14 LEGs) (B) using the Panglao DB 2021 database. (E) Representative Giemsa-Wright stained dissociated CFC replating (4th plating) MNX1 haemGx cells from 144h and 216h. Solid black arrows, mast cell precursors; dashed black arrows, mast cells.

CD41 and CD45, respectively. In agreement with recent observations from [Dias et al. \(2025\)](#), which highlight a critical balance between WNT-driven posterior and Nodal / TGF β anterior signals in patterning the midbody axis and endo-mesodermal derivatives, we observed that the combination of Activin A and CHIR99021 was critical for robust *Flk1/Kdr* activation and the efficient initiation of CD41 and CD45-marked hematopoietic specification. This is in contrast with the hematopoietic output of cardiogenic gastruloids reported by [Rossi et al. \(2022\)](#) which did not require Activin A. Although our strict dependence on Activin A to achieve initial activation of *Flk1* programs and specification of a vascular network and hematopoietic programs precluded a side-by-side comparison with the Rossi protocol ([Rossi et al., 2022](#)), we note their apparent lower efficiency in blood formation, at least at the comparable timepoints of 144 and 168h, and their bias towards erythroid formation, reflecting a putative YS, and potentially primitive, nature. This bias may also be captured in the reported spleen colony formation; however, given the limited characterisation of the colonies in the manuscript, which does not appraise origin, morphology or cellularity, the nature of the spleen engraftment remains speculative. Importantly, standard gastruloid protocols with only CHIR99021 pulse and no additional cytokines ([Beccari et al., 2018](#)) can capture a transcriptional erythrogenic wave.

Despite low frequency engraftment of adult hematopoietic tissues by haemGx cells, we demonstrated donor origin of the engrafted cells by flow cytometry and PCR detection of a BFP transgene up to 8 weeks. Engraftment progressed from erythroid-biased at 4 weeks to myeloid-lymphoid at 8 weeks, as observed in progenitor engraftment from adult cells ([Chun et al., 2024](#)). Critically, successful engraftment required implantation of undissociated haemGx for further maturation in a supportive microenvironment, in this case provided by the adrenal gland ([Schyrr et al., 2023](#)). In this study, we did not dissect the relative contribution of the continued maturation in 3D vs the influence of the niche for implantation, as we opted for performing the transplantation as to maximise both. Future experiments will be necessary to understand possible extrinsic niche contributions, including the sympathetic nervous system we sought to include ([Kapeni et al., 2022](#); [Fitch et al., 2012](#)), and how they can be incorporated in the haemGx culture by extension and/or further modification of the protocol. Although we observed continued growth and maturation of the haemGx *in situ* after implantation, this was not the case in all animals with engraftment. On the one hand, this dissociation excludes the likelihood of the engraftment resulting from teratoma formation from the gastruloid itself. On the other hand, it raises the possibility that the inefficiency of the engraftment is the consequence of a low frequency of cells within each haemGx with HSPC properties, which we can seek to resolve by upscaling of the haemGx culture or further expansion of dissociated HSPC ([Ng et al., 2024](#)).

Although the system is shy of achieving HSC maturation *in situ*, it can mature, to produce low-level engrafting multilineage progenitors, and potentially HSCs. This suggests that access to the relevant hematopoietic molecular programs is preserved. Incipient specification of non-hematopoietic tissues shown to support HSC emergence, including PDGFRA⁺ mesenchymal ([Chandranathan et al., 2022](#)) and neuro-mesenchymal ([Miladinovic et al., 2024](#)) cells, and neural-crest derivatives ([Kapeni et al., 2022](#); [Fitch et al., 2012](#)), occurs in sync with development of AGM-like hematopoiesis, to indicate the presence of cues that with further maturation may facilitate HSPC specification at the appropriate developmental time. Interestingly, through extending the haemGx culture, and in addition to AGM-like hematopoiesis, we also captured the potential to form renal tubular structures upon *in vivo* maturation, suggesting induction of lateral plate and intermediate mesoderm from which the AGM region derives, and putative reconstitution of the intra-embryonic hematopoietic niche.

Establishment of a tractable *in vitro* model of hematopoietic development which faithfully recapitulates time-dependent emergence of hematopoietic progenitors in their physiological niche would propel translational applications by enabling high-throughput studies of HSC emergence. Most ESC-based models of blood specification do not achieve concomitant recapitulation of the surrounding microenvironment. Protocols typically remove ESC-derived hemogenic cells from an

early embryonic-like niche to direct differentiation through genetic manipulation (Sugimura et al., 2017) and/or empirically-determined growth factor addition (Lis et al., 2017; Vo and Daley, 2015; Fowler et al., 2024; Nafria et al., 2020; Sroczynska et al., 2009; Piau et al., 2023), with minimal probing of intermediate steps. When hematopoietic waves have been specifically inspected, emergence of YS and AGM-like stages were conflated in time (Pearson et al., 2015; Murry and Keller, 2008), suggesting a bias towards intrinsic hematopoietic cell potential, at the expense of cell fate modulation by time-coherent extrinsic physiological cues, affecting the nature of cellular output. The present haemGx system, in contrast, successfully deconvolutes YS-like and AGM-like stages of blood production, including capturing differences in the HE generation, suggesting a level of organisation amenable to further development into an integrated hemogenic niche. Indeed, projections to scRNA-seq datasets of *in vivo* murine hematopoiesis place our endothelial and hematopoietic outputs at coherent temporal windows.

The haemGx protocol could be further developed to include more supportive structures of tissue organisation, as non-hematopoietic gastruloid models have benefitted from mechanical, or chemo-mechanical, support of artificial ECM embedding (Veenvliet et al., 2020; van den Brink, Susanne C et al., 2020; Dijkhuis et al., 2023), to match temporal coherence and tissue polarisation, with detailed tissue architecture illustrated by somite formation. Prior to HSC emergence in the AGM, critical signals such as SCF and Shh, which we have delivered to the cultures, are secreted by different cells on opposite dorsal-ventral locations (Souilhol et al., 2016), indicating the potential relevance of tissue organisation to improve and potentially extend hematopoietic formation. Moreover, HSPC formation in the AGM is facilitated by circulatory shear stress (North et al., 2009; Diaz et al., 2015; Lundin et al., 2020), suggesting that introduction of flow may improve hemato-endothelial organization and boost blood production.

We attested the utility of haemGx's to interrogate the biology of leukemias with developmental origins, thanks to a nearly full recapitulation of embryonic hemogenic specification, including critical early stages of HE production and EHT to EMP and MLP/MPPs. We modelled *MNX1*-r AML, a poorly characterized form of leukemia exclusive to the infant age group (Ragusa et al., 2023). Overexpression of *MNX1* in the haemGx model propagated cells bearing transcriptional resemblance to patient *MNX1*-r AML, overall positioning the haemGx as an attractive tractable model in which to study developmental leukemias. We showed that *MNX1*-OE transiently expanded a C-Kit⁺ cell with hemato-endothelial characteristics at the YS-like stage, and could selectively sustain the phenotype through serial replating of CFC assays. Replating CFCs could be initiated beyond the YS-like stage, albeit at the relative expense of an early differentiation state and of the similarity to patient transcriptome, putatively defining a window of susceptibility to *MNX1*-driven transformation.

Importantly, the haemGx model was able to resolve the cellularity and transcriptional features inferred from patient transcriptomes, with the advantage of a higher resolution in dissecting the contribution of cell-of-origin in the appropriate developmental context. A recent paper using *MNX1*-OE has shown that *MNX1* can initiate a transplantable leukemia in immunodeficient mice by targeting FL, but not BM, cells (Waraky et al., 2024). This suggests that ectopic expression of *MNX1* transforms an embryonic cell, rather than depends on an embryonic niche. It also suggests that *MNX1*-OE can cause leukemia as a single genetic hit, although it cannot be excluded that other mutations are required for, and may indeed have contributed to, full transformation *in vivo*, as observed in a recent report confirming the *in utero* origin of the leukemia (Bousquets-Muñoz et al., 2024). However, the mouse model does not address the nature of the embryonic cell targeted by *MNX1*-OE. Like the *MNX1*-OE haemGx cells selected through serial replating, *MNX1*-OE mouse leukemia are also C-Kit⁺ (Waraky et al., 2024). Interestingly, clusters of genes up-regulated in the mouse *MNX1*-OE leukemia (Fig. 7S1D – k1, 2, 5, 6, and 9) do not robustly separate *MNX1*-r from other infAML by GSEA analysis (Fig. 7S1E), and have diverged more from the patient data in their cell-type signature analysis than the original *MNX1*-OE cells in the FL (Fig. 7S1F), putatively selecting a more generic program of leukemia transformation. With the resolution provided by the

haemGx model, we propose that the divergence from *MNX1*-r infAML signatures increases from MNX1-OE haemGx 144h, to 216h to FL, positioning the origin of the AML at the YS stage, likely at the HE-to-EMP transition. This cell may persist in the haemGx culture system to the 216h timepoint, and migrate to the FL in the embryo, eventually extinguishing through differentiation, as suggested by the MNX1-OE haemGx CFC assays. Persistence of the original MNX1-OE target cell will determine the sensitivity to MNX1-OE, and can be addressed in the haemGx system through lineage tracing. Whether additional mutations, namely a reported association between *MNX1*-r and tri(19) (Espersen et al., 2018 [DOI](#); Ragusa et al., 2023 [DOI](#)), contribute to prolonging the undifferentiated state of the candidate YS target cell with increased probability of transformation, or otherwise expand the targeted cell in a more classical second-hit fashion, can potentially also be screened in the haemGx system, in combination with transplantation experiments.

Over the last decade, the effort to achieve robust hematopoietic production from PSC, mouse and human, has been centred on the generation of HSC, a critical translational need (Chang et al., 2023 [DOI](#); Dijkhuis et al., 2023 [DOI](#); Fidanza and Forrester, 2021 [DOI](#)). There are good indications that this is achievable, both with (Vo and Daley, 2015 [DOI](#); Dang et al., 2002 [DOI](#)) and without (Piau et al., 2023 [DOI](#)) genetic manipulation. However, success remains episodic, mainly because mechanistic understanding of processes underpinning blood generation during development is limiting, exacerbated by limiting access to human embryonic and fetal material. On-going optimization of gastruloid models and the success at integration of organoid cultures in microfluidic systems (Saorin, Caligiuri and Rizzolio, 2023 [DOI](#); Quintard et al., 2024 [DOI](#)) will allow us to further optimize haemGx culture. In conclusion, the haemGx represents a faithful *in vitro* reference to replace and complement embryo studies, holding promise to understand the biology of healthy and malignant developmental hematopoiesis and advance blood production for clinical use.

Experimental procedures

Materials

	SOURCE	IDENTIFIER
Antibodies		
Anti-mouse c-Kit (CD117) clone 2B8 - APC-Cy7	Biolegend	Cat. #105825
Anti-mouse CD41 clone MWRReg30 - PE-Dazzle594	Biolegend	Cat. #133935
Anti-mouse CD45 clone 30F11 - APC	Biolegend	Cat. #103111
Anti-mouse CD144 (VE-cadherin) clone BV13 - APC	Biolegend	Cat. #138011
Anti-mouse CD43 clone S11 - PE	Biolegend	Cat. #143205
Anti-mouse Ter119 - PC7	Biolegend	Cat. #116223
Anti-mouse CD31 - biotin	BDPharmingen	Cat. #553371
Anti-mouse CD34 - APC	Biolegend	Cat. #128611
CD11b anti-mouse/human clone M1/70 - biotin	Biolegend	Cat. #101203
Mouse Anti-Mouse CD45.2 clone 104 - PE	BD Biosciences	Cat. #560695
Rat Anti-Mouse CD31 clone MEC13.3 - biotin	BD Biosciences	Cat. #553371
Goat Anti-Mouse c-Kit/CD117	R&D Systems	Cat. #AF1356SP
Bacterial and virus strains		
<i>E. coli</i> : NEB 5-alpha Competent <i>E. coli</i>	New England Biolabs (NEB)	Cat. #C2987H
Chemicals, peptides, and recombinant proteins		
Murine LIF	Peprotech	Cat. #250-02
StemMACS PD0325901	Miltenyi Biotec	Cat. #130-106-5411
Chiron (CHIR99021)	Biogems	Cat. #2520691
Activin A PLUS	QKine	Cat. #Qk005
Murine VEGF 165	Peprotech	Cat. #450-32
Murine FGF-basic	Peprotech	Cat. #450-33
Murine Sonic Hedgehog (Shh)	Peprotech	Cat. #315-22
Murine TPO	Peprotech	Cat. #315-14
Murine Flt3-Ligand	Peprotech	Cat. #250-31L
Murine SCF	Peprotech	Cat. #250-03
N2B27 medium (NDiff 227)	Takara Bio	Cat. #Y40002
DMEM/F-12	Gibco	Cat. #11320033
Neurobasal medium	Gibco	Cat. #21103049
N-2 Supplement	Gibco	Cat. #17502048
B-27 Supplement	Gibco	Cat. #17504044
Critical commercial assays		
Mouse Methylcellulose Complete Media	R&D Systems	Cat. #HSC007
Deposited data		
Single cell RNA-seq of hemogenic gastruloids at timepoints	This paper	ArrayExpress: E-MTAB-12148
Bulk RNA-seq of hemogenic gastruloids with MNX1 overexpression at 144h and 216 h	This paper	ArrayExpress: E-MTAB-12173
Experimental models: Cell lines		
<i>Mus musculus</i> : Flk-1-GFP mouse embryonic stem cell line	Alfonso Martinez Arias Lab	Jakobsson <i>et al.</i> , 2010
<i>Mus musculus</i> : Sox17-GFP mouse embryonic stem cell line	Alfonso Martinez Arias Lab	Niakan <i>et al.</i> , 2010
<i>Mus musculus</i> : T/Bra::GFP mouse embryonic stem cell line	Alfonso Martinez Arias Lab	Fehling <i>et al.</i> , 2003
<i>Mus musculus</i> : E14Tg2a mouse embryonic stem cell line	MMRRC, University of California Davis, US	Hooper <i>et al.</i> , 1987

<i>Mus musculus</i> : Flk-1-GFP::Rosa26-BFP mouse embryonic stem cell line	This paper	n/a
<i>Homo sapiens</i> : HEK293T	ATCC	CRL-3216
Oligonucleotides		
MXN1 (forward 5-GTTCAAGCTCAACAAGTACC-3; reverse 5- GGTTCTGGAACCAAATCTTC-3)	Gulino <i>et al.</i> , 2021	n/a
Ppia (forward 5- TTACCCATCAAACCATTCTTCTG-3; reverse 5- AACCCAAAGAACTTCAGTGAGAGC-3)	Moris <i>et al.</i> , 2018	n/a
BFP (forward 5-GCACCGTGGACAACCATCACTT-3; reverse 5-CAGTTTGCTAGGGAGGTCGC-3)	This paper	n/a
Recombinant DNA		
pWPT-LSSmOrange-PQR	Ghevaert Lab, WT-MRC Cambridge Stem Cell Institute, UK	Dalby <i>et al.</i> , 2018
pWPT-LSSmOrange-PQR-MNX1	Cloned by Biomatik Corporation (Kitchener, Canada)	n/a
pcdna3-rosa26-rosa26-bfp	This paper	n/a
Software and algorithms		
GSEA software v4.2.3	Subramanian <i>et al.</i> , 2005	https://www.gsea-msigdb.org/gsea/index.jsp
ImageJ	Schneider <i>et al.</i> , 2012	https://imagej.nih.gov/ij/
Bowtie2	Langmead and Salzberg, 2012	http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
Samtools	Li <i>et al.</i> , 2009	http://samtools.sourceforge.net/
Tophat2	Kim <i>et al.</i> , 2013	https://ccb.jhu.edu/software/tophat/index.shtml
HTseq	Anders <i>et al.</i> , 2015	https://htseq.readthedocs.io/en/master/
DEseq2	Love <i>et al.</i> , 2014	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
ExpressAnalyst	n/a	https://www.expressanalyst.ca/
PANTHER	Thomas <i>et al.</i> , 2022	http://www.pantherdb.org/
EnrichR	Chen <i>et al.</i> , 2013; Kuleshov <i>et al.</i> , 2016; Xie <i>et al.</i> , 2021	https://maayanlab.cloud/Enrichr/
Panglao BP	Franzen <i>et al.</i> , 2019	https://panglaodb.se
TARGET (Therapeutically Applicable Research to Generate Effective Treatments) Database	https://ocg.cancer.gov/programs/target https://xenabrowser.net/	GDC TARGET-AML
Single-cell RNA sequencing analysis scripts	DockerHub: dsblab/single_cell_analysis:0.5	https://github.com/dsblab/blood_gastruloids
Other		
CELLSTAR cell-repellent cell culture plate, 96 well, U-bottom	Greiner (Bio-One)	Cat. #650970

Methods

Cell culture

Flk-1-GFP (Jakobsson *et al.*, 2010 [DOI](#)), Flk-1-GFP::Rosa26-BFP, Sox17-GFP (Niakan *et al.*, 2010 [DOI](#)), T/Bra::GFP (Fehling *et al.*, 2003 [DOI](#)), and E14Tg2A (Hooper *et al.*, 1987 [DOI](#)) mouse embryonic stem cells (mES) lines were cultured in ES+LIF medium in gelatinized flasks (0.1% gelatin) with daily

medium change, as previously described (Turner et al., 2017 [DOI](#)). The ES+LIF medium contained 500 ml Glasgow MEM BHK-21 (Gibco), 50 ml of fetal bovine serum (FBS, Embryonic Stem Cells tested, biosera, Nuaille, France), 5 ml GlutaMAX supplement (Gibco), 5 ml MEM Non-Essential Amino Acids (Gibco), 5 ml Sodium pyruvate solution (100 mM, Gibco), and 1 ml 2-Mercaptoethanol (50mM, Gibco). Murine LIF (Peprotech) was added at 1000 U/ml. HEK293T cells were grown in Dulbecco's modified Eagle medium (Gibco) supplemented with 10% FBS (Gibco) and 1% penicillin/streptomycin (Gibco). All cultures were kept at 37°C and 5% CO₂.

Hemogenic gastruloids assembly, culture, and dissociation

mES were maintained in ES+LIF medium and transferred to 2i+LIF (containing Chiron and MEK inhibitor PD03) for 24 hours prior to the assembly into gastruloids. 250 untransduced mES cells (or 400 for GX-MNX1 and GX-EV) were seeded in each well of a U-bottom, cell-repellent 96-well plate (Greiner Bio-One, Stonehouse, UK) in 40 µl of N2B27 medium [Takara Bio or home-made as Cold Spring Harbour Protocols 'N2B27 Medium' (2017)]. The plate was centrifuged at 750 rpm for 2 minutes to promote deposition and aggregation of the cells and was then incubated at 37°C, 5% CO₂ for 48 hours. After 48 hours, 150 µl of N2B27 medium supplemented with 100 ng/ml Activin A (QKine, Cambridge, UK) and 3 µM chiron (Peprotech) was added to each well. At 72 hours, 150 µl of medium were removed, without disrupting the gastruloids in the wells. 100 µl of N2B27 with 5 ng/ml of Vegf and Fgf2 each (Peprotech) were added to each well. From 72 h to 144 h, each day 100 µl of medium were removed and replaced with N2B27 + Vegf + Fgf2. At 144 h, the medium was further supplemented with Shh at 20 ng/ml. From 168 h to 216 h, the medium was N2B27 + 5 ng/ml Vegf, plus 20 ng/ml mTpo, 100 ng/ml mFlt3l, and 100 ng/ml mScf (Peprotech). To dissociate cells from the gastruloid structures, medium was removed and individual gastruloids were collected using a pipette and precipitated at the bottom of a microcentrifuge tube. The remaining medium was aspirated and the bulk of gastruloids was washed in PBS. 50 µl of TrypLE express was added to pelleted gastruloids to be incubated at 37°C for 2 minutes to dissociate cells.

Animals and implantation

Female athymic nude-Foxn1^{nu} (*nu/nu*) mice (Envigo, Bresso, Italy) were housed under pathogen-free conditions. In accordance with the "3Rs policy", experiments were reviewed and approved by the Animal Welfare Body (OPBA) of IRCCS Ospedale Policlinico San Martino and by the Italian Ministry of Health (n. 883/2020-PR).

Intact haemGx (3 per mouse in 10 µL of PBS) were inoculated in the adrenal gland of five-week-old mice. Briefly, mice were anesthetized with a mixture of xylazine (ROMPUN, Bayer) and ketamine (LOBOTOR, Acme S.r.l.) and injected with haemGx, after laparotomy, in the capsule of the left adrenal gland, as previously described (Brignole et al., 2023 [DOI](#); Pastorino et al., 2003 [DOI](#)). All mice survived surgery. Control mice (CTR) received vehicle only. Mice body weight and general physical conditions were recorded daily. One month after haemGx inoculation, mice were sacrificed and spleen and bone marrow (BM) from femurs collected. BM cells and spleens, mechanically reduced to cell suspension, were stored at -80°C until use. Adrenal glands were paraffin-embedded for subsequent IHC studies.

Methylcellulose colony forming assays (CFC)

Disassembled gastruloids cells were plated on Mouse Methylcellulose Complete Media (R&D Systems). Cells were first suspended in Iscove's Modified Dulbecco's Medium (IMDM, Gibco) with 20% FBS (Gibco) before addition to the methylcellulose medium. Cells were plated in duplicate 35 mm dishes with 2×10⁵ cells/plate. Plates were incubated at 37°C and 5% CO₂ for 10 days, when colonies were scored. For serial replating experiments, cells in methylcellulose were collected and washed in phosphate buffer saline (PBS) to achieve single-cell suspensions and replated as described above. Dissociated gastruloid cells from CFC were collected as cell suspensions in PBS

and centrifuged onto slides (Shandon Cytospin 2 Cytocentrifuge) at 350rpm for 5-7min, with a PBS-only pre-spin at 1000rpm, 2min. Slides were stained in Giemsa-Wright stain for 30 seconds followed by addition of phosphate buffer pH 6.5 for 5 min.

Immunofluorescence staining

Immunostaining of whole gastruloids was performed as described before (Baillie-Johnson et al., 2015). Briefly, gastruloids were fixed in 4% paraformaldehyde (PFA) dissolved in PBS for 4 hours at 4°C on orbital shaking and permeabilised in PBSFT (10% FBS and 0.2% Triton X-100), followed by one hour blocking in PBSFT at 4°C on orbital shaking. Antibody dilutions were made in PBSFT at 1:200 for primary and 1:500 for secondary antibodies. Antibody incubations were performed overnight at 4°C on orbital shaking, and subjected to optical clearing overnight in ScaleS4 clearing solution. Individual gastruloids were then mounted on glass coverslips by pipetting as droplets in ScaleS4 and DAPI nuclear stain.

Histology and H&E staining

Samples were fixed with 10% formalin at room temperature for 48 hours followed by an ethanol series of ethanol and paraffin; ethanol 70% (90 minutes), ethanol 80% (120 minutes), ethanol 95% (180 minutes), 3 times ethanol 100% of 120, 120 and 180 minutes, 50% ethanol/paraffin (3×120 minutes) and 3 times paraffin (60, 90 and 120 minutes) before embedding in histological cassettes. Blocks were sectioned with a HistoCore BIOCUT Microtome (Leica) at 10 mm and slides were stained with Haematoxylin/Eosin.

Imaging

Images of cultured gastruloids and CFC plates were captured using the Cytation 5 Cell Imaging Multi-Mode Reader (Biotek) plate reader using bright field and FITC channels. ImageJ (Schneider et al., 2012) was used for gastruloid size quantification. Confocal microscopy was performed on LSM700 on a Zeiss Axiovert 200 M with Zeiss EC Plan-Neofluar 10x/0.30 M27 and Zeiss LD Plan-Neofluar 20x/0.4 M27 objective lens. Tissue sections were imaged with a THUNDER Imager 3D microscope system (Leica).

Lentiviral vector packaging and transduction

The lentiviral overexpression vector pWPT-LSSmOrange-PQR was used to clone the *MX1* gene cDNA. The viral packaging vectors pCMV and pMD2.G, described in Pina et al. (2008) [\[14\]](#), were used to assemble lentiviral particles using HEK293T cells via transfection using TurboFect Reagent (Invitrogen). Transduction of mES cells was performed overnight by addition of lentivirus to culture medium and washed the following day (Moris et al., 2018) [\[15\]](#). Transduced cells were sorted for positivity to LSSmOrange.

Flow cytometry

Surface cell marker analysis was performed by staining using the antibodies listed in Key Resources Table. Disassembled gastruloids cells were resuspended in PBS, 2% FBS and 0.5 mM EDTA and stained at a dilution of 1:100 for primary antibodies for 20 minutes at 4°C. When indicated, streptavidin was added at a dilution of 1:200. Analysis was performed on ACEA Novocyte (Agilent) or AttuneNxt (Thermo) analyzers, using the respective software packages. Cell sorting was performed using a CS&T calibrated BDFACS Aria III system (488nm 40 mW, 633nm 20 mW, 405nm 30 mW, and 561nm 50 mW), set with the 100µm nozzle at 20PSI and a 4-way purity mask or, in the case of haemGx adrenal implants, a Beckman Coulter CytoFlex SRT (488nm 50 mW, 633nm 100 mW, 405nm 90 mW, and 561nm 30 mW), set with 100µm nozzle at 15PSI, also using 4-way purity. Single-cell deposition in 96-well plates was performed using single-cell sorting mode.

Intact cells were gated on FSC-A vs SSC-A plot, followed by doublet exclusion on FSC-A vs FSC-H and SSC-A vs SSC-H, prior to gating on fluorescent parameters for the markers described in the results.

Single cell RNA sequencing of time-resolved hemogenic gastruloids

Gastruloids were collected at different timepoints of the protocol, disassembled and FACS deposited into 96-well plates, either as unsorted (global) or sorted by CD45, CD41 and C-Kit/Sca1 markers (sorted). RNA from the cells were extracted from single cells using Smart-seq2 technology at 500Kb and depth of 151 bases. Sequencing reads were quality-checked using FastQC (v0.11.9). We trimmed the samples using trimGalore! (0.6.6) with a cutoff of 30, clipping 15 base pairs and retaining reads of more than 100 bases. Alignment was performed on STAR (2.7.8a) and Mus musculus annotations from GENCODE vM26. Aligned BAM files were annotated using featureCounts (v2.0.1) and count matrices were computed in python by directly accessing the BAM files with gtfparse packages and collapsing lanes.

For quality control, based on the histogram of counts and multimodality distributions, we set a minimum count threshold of 200000 counts, a minimum threshold of 1000 expressed genes, and a maximum threshold of 20% mitochondrial fraction per cell. We performed the same procedure of the second dataset with a more restrictive threshold of 400000 counts and similar expressed genes and mitochondrial thresholds. Cells that did not pass the quality control metrics were omitted from analysis. We normalized the cells to the mean count number per dataset and applied a plus-one-log transformation of the data before proceeding to the downstream analysis.

Dimensionality reduction was performed on feature selection of the gene space using the function `scanpy.highly_varying_genes` with default parameters. Selection of principal components in principal component analysis (PCA) was performed by heuristic elbow method. Nearest neighbor analysis was constructed by KNN graphs using a correlation metric and 10 nearest neighbors. Data was projected for low dimensional visualization using the UMAP algorithm with default parameters as implemented in `scanpy.tl.umap`. We used the leiden algorithm as implemented in `scanpy.tl.leiden` to partition the data into clusters. In order to assess the election of the resolution parameter we used Newman-Girvan modularity as a metric of clustering quality. Differential expressed genes were computed comparing each cluster against the rest using the Wilcoxon test with Benjamini-Hochberg correction.

Annotation and projection to additional datasets

Raw counts matrices were downloaded and processed following the same pipeline as described above for scRNA-seq of gastruloids to generate UMAP and clustering. We implemented the `scmap` algorithm to compare our scRNA sequencing of gastruloids with the available datasets. We reduced the dimensionality of the space by selecting highly varying genes from the annotated dataset. Then, we constructed a KNN classifier with correlation metric and computed the nearest neighbors of the target data. If neighbors with correlation metrics below 0.7 default standards, the projected cells were not projected onto any cell from the annotated dataset. To visualize the cells over the UMAP plots of the other datasets, we constructed a KNN regressor with three neighbors and a correlation metric. Confusion matrices were used to visualize the overlap between gastruloid clusters and the embryo dataset assigned clusters using `scmap`.

Bulk RNA sequencing

Total RNA was extracted from disassembled gastruloid cells at 216 hours. Sequencing was performed on NovaSeq PE150 platform, at 20M paired-end reads per sample. Tophat2 with Bowtie2 were used to map paired-end reads to the reference Mus musculus genome build GRCm39. GENCODE Release M30 (Frankish et al., 2019 [DOI](#)) was used as the reference mouse genome annotation. Aligned reads were filtered by quality using samtools (Li, H. et al., 2009 [DOI](#)) with a

minimum threshold set at 30 (q30). Transcript assembly and quantification was achieved using htseq (Putri et al., 2022). Differential expression between sample and control was performed by collapsing technical replicates for each condition using Deseq2 (Love, Huber and Anders, 2014) in R environment (Deseq2 library v 1.32.0). The differential expression was expressed in the form of log2 fold change and filtered by false discovery rate (FDR) of 0.05.

Gene list enrichment analyses

Gene ontology (GO) analysis was performed in ExpressAnalyst (available at www.expressanalyst.ca) using the PANTHER or GO Biological Process (BP) repository. GO terms and pathways were filtered by false discovery rate (FDR) with a cut-off of ≤ 0.1 for meaningful association. EnrichR (Xie et al., 2021; Kuleshov et al., 2016; Chen, E. Y. et al., 2013) was used for cell type analysis using the Panglao DB (Franzén, Gan and Björkegren, 2019) databases using a FDR threshold of ≤ 0.1 or p value ≤ 0.01 , where specified, as well as transcription factor (TF) binding site enrichment using the ENCODE and ChEA Consensus TFs from ChIP database.

Gene Set Enrichment Analysis (GSEA)

Custom gene signatures were used as gene sets for GSEA analysis (Subramanian et al., 2005) on the GSEA software v4.2.3 on RNA sequencing expression values in counts units. GSEA was ran in 10000 permutations on gene set using the weighted Signal2Noise metric. Enrichment metrics are shown as normalized enrichment score (NES) and filtered by FDR ≤ 0.05 . Leading edge genes (LEGs) are genes with a “Yes” values for core enrichment. For AML patient analysis, clinical phenotype and expression data (in counts units) were extracted from the GDC TARGET-AML cohorts in the Therapeutically Applicable Research to Generate Effective Treatments project (TARGET, <https://ocg.cancer.gov/programs/target>), downloaded from the University of California Santa Cruz (UCSC) Xena public repository (last accessed 31st August 2022). Patient samples were selected according to the reported karyotype to include t(7;12), inv(16), MLL, normal karyotype, and t(8;21). GSEA was performed comparing RNA sequencing counts of t(7;12) samples against pooled AML subtypes (inv(16), MLL, normal karyotype and t(8;21)) as “other AML”.

Real-time polymerase chain reaction (qPCR)

Extracted RNA was reverse-transcribed into complementary DNA (cDNA) using High-Capacity RNA-to-cDNA Kit (Applied Biosystems). QPCR was performed using FastGene 2x IC Green Universal qPCR Mix (Nippon Genetics, Duren, Germany) using primers for human *MX1* (forward 5-GTTCAAGCTCAACAAGTACC-3; reverse 5-GGTTCTGGAACCAATCTTC-3) (Gulino et al., 2021) and *Ppia* for endogenous control (forward 5-TTACCCATCAAACCATTCCTTCTG-3; reverse 5-AACCCAAAGAACTTCAGTGAGAGC-3) (Moris et al., 2018), or Taqman gene expression assays (Mm00501741_m1 *Myb*; Mm00439364_m1 *Hoxa9*; Mm00723157_m1 *Hlf*; Mm01213404_m1 *Runx1*; Mm00444619_m1 *Dll4*; Mm00491303_m1 *Mecom*; Mm02342430_g1 *Ppia*) using Taqman Gene Expression Mastermix (Applied Biosystems). Differential gene expression was calculated using the delta delta Ct ($\Delta\Delta Ct$) method.

Polymerase chain reaction (PCR)

Genomic DNA was extracted using Monarch Genomic DNA Purification Kit (New England Biolabs, Hitchin, UK) using manufacturer’s instructions. PCR on genomic DNA was performed using Phire PCR Master Mix (Thermo Scientific) to amplify the BFP locus using forward primer 5’-GCACCGTGGACAACCATCACTT-3’ and reverse primer 5’-CAGTTTGCTAGGGAGGTCGC-3’.

Statistical analysis

Experiments were performed at least in triplicates, unless specified otherwise. Data are plotted to include standard deviation (+/-SD) between replicates. Statistical significance was set at a threshold of p value < 0.05. Statistical analysis was performed in R environment (version 4.1.3) or using GraphPad Prism 8.0 software and is detailed in respective figure legends.

Data availability

Raw data as well as processed count matrices and post-processed files from single-cell RNA-seq for the time-resolved data is available at E-MTAB-12148. Bulk RNA-seq of MNX1 overexpressing gastruloids is available at Array Express with accession code E-MTAB-12173. The post-processing was performed in Python on DockerHub: dsblab/single_cell_analysis:0.5. Scripts are available in https://github.com/dsb-lab/blood_gastruloids and Zenodo (<https://doi.org/10.5281/zenodo.7053423>). The results published here are partly based upon data generated by the Therapeutically Applicable Research to Generate Effective Treatments (TARGET) (<https://ocg.cancer.gov/programs/target>) initiative, of the Acute Myeloid Leukemia (AML) cohort GDC TARGET-AML. The data used for this analysis are available at <https://portal.gdc.cancer.gov/projects> and <https://xenabrowser.net/>.

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Additional information

Author contributions

Conceptualization: CP, AMA; Methodology: CWS, DR, GTC, FP, SvdB, MC,MP, KRK, VH-H, AB, JGO, AMA, CP; Software: GTC, JGO; Validation: CWS, DR, AJ, YC, SvdB, CP; Investigation: CWS, DR, GTC, FP, AJ, YC, LD, CB, G-AI, CP; Formal Analysis: GTC, DR, JGO, CP; Resources: GTC, JC, MP, JGO, AMA; Data curation: DR, GTC, JGO; Writing – Original Draft: CP, DR, GTC; Writing – Review and editing: CP, DR, GTC, AB, JGO, AMA; Visualisation: DR, GTC, CWS, CP; Supervision: CP; Project administration: CP, AMA, JGO; Funding acquisition: CP, AMA, AB.

Additional files

Supplemental File S1. Cluster classifier gene lists for each cluster by differential gene expression analysis of scRNA-seq of haemGx (all conditions), obtained by Wilcoxon rank test of each cluster against all other clusters. [↗](#)

Supplemental File S2. Cell type enrichment analysis of cluster classifier genes, showing the inferred identity of clusters of scRNA-seq of haemGx (all conditions). Classifier genes (in Supplemental File S1) were obtained by differential expression in comparison to all other clusters and subjected to enrichment analysis using the PanglaoDB repository (Franzén, Gan and Björkegren, 2019). [↗](#)

Supplemental File S3. Summary of projections of scRNA-seq of haemGx cells onto available datasets by Hou et al. (2020), Zhu et al. (2020), and Thambyrajah et al. (2024), showing the identity of each haemGx cell projecting to annotated clusters of the respective studies. [↗](#)

Supplemental File S4. List of differentially expressed genes between haemGx-MNX1 and haemGx-MNX1 at 144h and 216h compared to empty vector (EV) control. [↗](#)

Supplementary Figures [↗](#)

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Reviewer #1 (Public review):

Summary

This manuscript describes a haemogenic gastruloid system that the authors claim recapitulates early mouse embryonic development to produce sequential waves of yolk sac and AGM-like haematopoiesis, with spatial and temporal accuracy. The model claims to reproduce mouse development to 'beyond' the E9.0 stage and apply its use to the aetiology of infant leukaemia.

Strengths

Gastruloids models are useful systems for studying early embryonic development, recapitulating aspects of gastrulation, anteroposterior regionalisation and somitogenesis. Gastruloid models that specifically mimic particular regions of the embryo could provide insights into how these regions form during development.

Weaknesses

There are a couple of major issues with this manuscript that I feel need to be addressed.

Firstly, the authors acknowledge that the proportion of blood cells that are produced by their haemogenic gastruloid system is very low - there are fewer than 2% of either blood or endothelium produced. The authors argue however, that this is because they have developed a hematopoietic organoid that captures much more of the essence of the developing embryo and therefore has a broader tissue representation and a more relevant spatial representation.

In order to prosecute this argument, this reviewer needs to understand how the differentiation protocol achieves this end, ie what is notable about the combination of factors and other media components. Also, they need to know what the evidence is to support this claim, in other words, what are the tissues that make up the organoid and is it truly representative of what would be expected in a developing embryo over this time. Does it pass from epiblast to primitive streak and then to cells of the germ layers? And how do haemGxs at different times map onto the developing mouse embryo?

Secondly, the point is repeatedly made by the authors that the distinction between non-engrafting yolk sac hematopoiesis and AGM-like hematopoiesis from which repopulating HSCs first derive is not really possible without spatial cues. This is really not true. It has been shown by a number of investigators, and summarised in a recent review (Abuhantash et al 2021), that the expression of HOXA cluster genes - most prominently HOXA9 - clearly distinguishes AGM-derived, from yolk sac derived cells. In this manner, it is evident from the UMAP provided that there is no HOXA9 expressed in either endothelium or blood cells. This argues very strongly against the proposition that AGM-type hematopoiesis is generated. Indeed, given the duration of the organoid culture of only 9 days (216hrs), it would be highly unlikely that development would even reach the stage of AGM hematopoiesis (E11.5 in the mouse), even with a 1:1 concordance between embryonic time and in vitro differentiation. Finally, if there is recapitulation of the normal pattern of embryogenesis, it would be expected that there would be a prominent phase of yolk sac hematopoiesis antedating AGM-associated hematopoiesis, which should be observed in the haemGx.

I feel that these are major conceptual points that need to be addressed in this manuscript.

<https://doi.org/10.7554/eLife.102324.2.sa3>

Reviewer #2 (Public review):**Summary:**

In this manuscript, the authors describe the development of a new hemogenic gastruloid (hGX) system, which they claim recapitulates the sequential generation of various hematopoietic cell types. A key proposed advantage of this system is its ability to more faithfully model the spatiotemporal emergence of hematopoietic progenitors within a physiologically relevant niche, as compared to existing in vitro platforms. While the authors provide some initial characterisation and demonstrate the utility of the system in studying infant leukemia, the presented data are not fully conclusive and fall short of robustly supporting several of their key assertions.

Strengths:

The development of a novel in vitro system to model hematopoietic development is innovative and could potentially address important limitations of existing platforms.

Weaknesses:

The characterization of the hematopoietic progenitors generated by the hGX system is not fully convincing. The evidence supporting the emergence of late yolk sac (YS) progenitors, including lymphoid cells, and AGM-like pre-hematopoietic stem cells (pre-HSCs), is incomplete and relies heavily on transcriptomic profiling and a limited set of markers.

The identification of lymphoid or pre-HSC-like populations is primarily inferred from scRNA-seq data. The lack of robust functional validation (e.g., lymphoid differentiation assays or long-term repopulation experiments) significantly weakens the manuscript's main claims.

In the revised manuscript, the authors incorporate single-cell RNA-seq analyses indicating that their cells resemble AGM-derived endothelial-to-hematopoietic transition (EHT) populations. However, they do not test whether these cells might more closely resemble YS-derived EHT, which remains an unresolved and critical question. Additionally, the claim in line 263 that Cluster 8 (CD45⁺ cells at 192-216 h) expresses lymphoid markers is not clearly supported by the provided supplemental data (Supplemental File S1-S2).

While the authors respond that they did not claim to generate bona fide HSCs, they do state at the end of the Introduction (lines 116-121) that their system captures AGM hematopoiesis. The current data do not support this conclusion and instead suggest that the system recapitulates the generation of multipotent lymphoid progenitors (MLPs) akin to those found in the YS.

The engraftment data presented are not particularly convincing. It is unclear why the analysis was terminated at 8 weeks post-transplant, especially given that a minimum of 12 weeks is generally required to meaningfully assess the presence of pre-HSCs or bona fide HSCs with long-term repopulating potential.

Given the uncertainties discussed above, the interpretability of the MNX1 overexpression study is limited.

The authors could have more directly tested their claim of capturing multiple hematopoietic waves by performing kinetic analyses of colony-forming potential, with the expectation that more multipotent colonies would emerge at later time points. Additionally, isolating and characterizing the potential of hemogenic endothelium at different time points corresponding to the putative waves would have strengthened their conclusions. In the absence of such data, it remains unclear whether the system recapitulates sequential waves

of hematopoiesis or merely reflects the progressive maturation of cells originating from a single wave.

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Reviewer #3 (Public review):

The authors present a revised version of their manuscript (Ragusa et al.) describing a hemogenic gastruloid (haemGx) model, used to investigate stages of blood production in vitro and for modeling a rare type of infant leukemia. The revisions address several major concerns raised during the initial round of review, and new data have been provided that overall improve the clarity and rigour of the study. In particular, the additional flow cytometry, single-cell RNA-seq analyses, and benchmarking against in vivo datasets help, to some extent, to substantiate the claims of developmental relevance of haemGx to yolk sac (YS)- and AGM-like hematopoietic waves. Nonetheless, some issues remain, particularly regarding the claims of short-term engraftment, novelty of the model, and the extent to which AGM-like HSPC are truly captured.

Major Points:

- (1) The authors have clarified the novelty of their haemGx protocol relative to existing gastruloid models, including the importance of the Activin A pulse and protocol extension to 216h. Flow cytometry and scRNA-seq analyses support the emergence of endothelial and hematopoietic populations with dynamic marker expression. However, direct side-by-side comparisons with previously published protocols (e.g., Rossi et al., 2022) remain limited. The claim of "spatio-temporal accuracy" should be more cautiously phrased.
- (2) The characterization of the identity of the hematopoietic waves generated in the haemGx system has been improved in the revised manuscript. Flow cytometry analysis now includes CD31/CD34 co-expression in CD41+ and CD45+ subsets, and scRNA-seq re-clustering supports two hematopoietic waves with distinct marker sets (e.g., Gata2/Myb vs. Hoxa9/Ikzf1). Projection onto multiple embryonic reference datasets (Hou et al., Zhu et al., Thambyrajah et al.) is a valuable addition. The case for YS-like EMP and AGM-like HSPC precursors is reasonably made, though further functional distinctions (e.g., lineage output differences) would strengthen the claims.
- (3) The authors have now provided additional evidence for low-level engraftment following adrenal implantation of whole haemGx. Although technically demanding, this in vivo result remains marginal and should be interpreted with caution. Crucially, this still does not demonstrate HSC-level repopulation capacity. The revised manuscript has softened the claims accordingly, now referring to "progenitor" activity rather than "pre-HSC." We agree that this adjusted claim is more suitable, though the reproducibility of this experiment is still unclear.
- (4) The MNX1 overexpression experiments are generally convincing in showing early expansion of a putative HE-to-EMP-like population and transcriptional resemblance to MNX1-r AML. However, the evidence for transformation is still solely based on in vitro data and lacks any evidence of in vivo leukaemia engraftment. The ability to perturb the system would add translational value to the haemGx platform, although future studies are needed to better define transformation dynamics and leukemogenic progression.

<https://doi.org/10.7554/eLife.102324.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary

The authors describe a method for gastruloid formation using mouse embryonic stem cells (mESCs) to study YS and AGM-like hematopoietic differentiation. They characterise the gastruloids during nine days of differentiation using a number of techniques including flow cytometry and single-cell RNA sequencing. They compare their findings to a published data set derived from E10-11.5 mouse AGM. At d9, gastruloids were transplanted under the adrenal gland capsule of immunocompromised mice to look for the development of cells capable of engrafting the mouse bone marrow. The authors then applied the gastruloid protocol to study overexpression of Mnx1 which causes infant AML in humans.

In the introduction, the authors define their interpretation of the different waves of hematopoiesis that occur during development. 'The subsequent wave, known as definitive, produces: first, oligopotent erythro-myeloid progenitors (EMPs) in the YS (E8-E8.5); and later myelo-lymphoid progenitors (MLPs - E9.5-E10), multipotent progenitors (MPPs - E10-E11.5), and hematopoietic stem cells (HSCs - E10.5-E11.5), in the aorta-gonad-mesonephros (AGM) region of the embryo proper.' Herein they designate the yolk sac-derived wave of EMP hematopoiesis as definitive, according to convention, although paradoxically it does not develop from intra-embryonic mesoderm or give rise to HSCs.

Our definition of primitive and definitive waves is widely used in the field (e.g. PMID: 18204427; PMID: 28299650; PMID: 33681211). Definitive haematopoiesis, encompassing EMP, MLP, MPP and HSC, highlights their origin from haemogenic endothelium, generation of mature cells with adult characteristics from progenitors with multilineage potential and direct and indirect developmental contributions to the intra-embryonic and time-restricted generation of HSCs.

General comments

The authors make the following claims in the paper:

- (1) The development of a protocol for hemogenic gastruloids (hGx) that recapitulates YS and AGMlike waves of blood from HE.*
- (2) The protocol recapitulates both YS and EMP-MPP embryonic blood development 'with spatial and temporal accuracy'.*
- (3) The protocol generates HSC precursors capable of short-term engraftment in an adrenal niche.*
- (4) Overexpression of MNX1 in hGx transforms YS EMP to 'recapitulate patient transcriptional signatures'.*
- (5) hGx is a model to study normal and leukaemic embryonic hematopoiesis.*

There are major concerns with the manuscript. The statements and claims made by the authors are not supported by the data presented, data is overinterpreted, and the conclusions cannot be justified. Furthermore, the data is presented in a way that makes it difficult for the reader to follow the narrative, causing confusion. The authors have not discussed how their hGx compares to the previously published mouse embryoid body protocols used to model early development and hematopoiesis. Specific points

(1) It is claimed that HGxs capture cellularity and topography of developmental blood formation. The hGx protocol described in the manuscript is a modification of a previously published gastruloid protocol (Rossi et al 2022). The rationale for the protocol modifications is not fully explained or justified. There is a lack of novelty in the presented protocol as the only modifications appear to be the inclusion of Activin A and an extension of the differentiation period from 7 to 9 days of culture. No direct comparison has been made between the two versions of gastruloid differentiation to justify the changes.

The Reviewer paradoxically claims that the protocol is not novel and that it differs from a previous publication in at least 2 ways – the patterning pulse and the length of the protocol. Of these, the patterning pulse is key. As documented in Fig. 1S1, we cannot obtain Flk1-GFP expression in the absence of Activin A (Fig. 1S1A), and the concentration of Activin A scales activity of the Flk1 locus (Fig. 1S1B). Expression of Flk1 is a fundamental step in haemato-endothelial specification and, accordingly, we do not see CD41 or CD45+ cells in the absence of Activin A. Furthermore, these markers also titrate with the dose of Activin A (in Fig. 1S1B).

Also, in our hands, there is a clear time-dependent progression of marker expression, with sequential acquisition of CD41 and CD45, with the latter not detectable until 192h (Fig. 1C-D), another key difference relative to the Rossi et al (2022) protocol. We suggest, and present further evidence for in this rebuttal and the revised manuscript, that the 192h-timepoint captures the onset of AGM-like haematopoiesis. We have edited the manuscript to clarify the differences and novelty in our protocol (lines 132-143) and provided a more detailed comparison with the report from Rossi et al. (2022) in the Discussion (lines 574-586).

The inclusion of Activin A at high concentration at the beginning of differentiation would be expected to pattern endoderm rather than mesoderm. BMP signaling is required to induce Flk1+ mesoderm, even in the presence of Wnt.

Again, we call the Reviewer's attention to Fig. 1S1A which clearly shows that Activin A (with no BMP added) is required for induction of Flk1 expression, in the presence of Wnt. Activin A in combination with Wnt, is used in other protocols of haemato-endothelial differentiation from pluripotent cells, with no BMP added in the same step of patterning and differentiation (PMID: 39227582; PMID: 39223325). In the latter protocol, we also call the Reviewer's attention to the fact that a higher concentration of Activin A precludes the need for BMP4 addition. Finally, one of us has recently reported that Activin A, on its own, will induce Flk1, as well as other anterior mesodermal progenitors (<https://www.biorxiv.org/content/10.1101/2025.01.11.632562v1>). In addressing the Reviewer's concerns with the dose of Activin A used, we titrated its concentration against activation of Flk1, confirming optimal Flk1-GFP expression at the 100ng/ml dose used in the manuscript. We have included this data in the manuscript in Figure 1S1B.

FACS analysis of the hGx during differentiation is needed to demonstrate the co-expression of Flk1GFP and lineage markers such as CD34 to indicate patterning of endothelium from Flk1+ mesoderm. The FACS plots in Fig. 1 show C-Kit expression but very little VE-cadherin which suggests that CD34 is not induced. Early endoderm expresses C-Kit, CXCR4, and Epcam, but not CD34 which could account for the lack of vascular structures within the hGx as shown in Fig. 1E.

We were surprised by the Reviewer's comment that there are no endothelial structures in our haemogenic gastruloids. The presence of a Flk1-GFP+ network is visible in the GFP images in Fig. 1B, from 144h onwards, and is detailed in the revised Fig. 2A, which shows overlap between Flk1GFP and the endothelial marker CD31. In addition, our single-cell RNA-seq data, included in the manuscript, confirms the presence of endothelial cells with a developing

endothelial, including arterial, programme. This is now presented in the revised Fig. 3B-D of the manuscript, which updates a representation in the original manuscript. In contrast with the Reviewer's claims that no endothelial cells are formed, the data show that Kdr (Flk1)+ cells co-express Cdh5/VE-Cadherin and indeed Cd34, attesting to the presence of an endothelial programme. Arterial markers Efnb2, Flt1, and Dll4 are present. A full-blown programme, which also includes haemogenic markers including Sox17, Esam, Cd44 and Mecom is clear at early (144h) and, particularly at late (192h) timepoints in cells sorted on detection of surface C-Kit (Fig. 3B-E in the manuscript). To address the specific point by the Reviewer, we also document co-expression of Flk1-GFP, CD34 and/or CD31 by flow cytometry (Fig. 2S1A-B in the revised manuscript).

To summarise new and revised data in the manuscript in relation to this point:

Immunofluorescence staining showing the Flk1-GFP-defined vascular network in Figure 1E and co-expression of endothelial marker CD31 in Figure 2A. In text: lines 159-163; 178-180.

Flow cytometry analysis of co-expression of Flk1-GFP with CD31 and CD34 in Figure 2S1AD, including controls. In text: 180-187.

Real-time quantitative (q)PCR analysis showing time-dependent expression of haematoendothelial and arterial markers in Figure 2F (specifically Dll4 and Mecom). In text: 200-209.

An improved representation of our scRNA-seq data highlighting key haemato-endothelial markers in Figure 3B-D. In text: 268-304

(2) The protocol has been incompletely characterised, and the authors have not shown how they can distinguish between either wave of Yolk Sac (YS) hematopoiesis (primitive erythroid/macrophage and erythro-myeloid EMP) or between YS and intraembryonic Aorta-Gonad-Mesonephros (AGM) hematopoiesis. No evidence of germ layer specification has been presented to confirm gastruloid formation, organisation, and functional ability to mimic early development. Furthermore, differentiation of YS primitive and YS EMP stages of development in vitro should result in the efficient generation of CD34+ endothelial and hematopoietic cells. There is no flow cytometry analysis showing the kinetics of CD34 cell generation during differentiation. Benchmarking the hGx against developing mouse YS and embryo data sets would be an important verification.

The Reviewer is correct that we have not provided detailed characterisation of the different germ layers, as this was not the focus of the study. In that context, we were surprised by the earlier comment assuming co-expression of C-Kit, Cxcr4 and Epcam, which we did not show, while overlooking the endothelial programme reiterated above, which we have presented. Given our focus on haemato-endothelial specification, we have started the single-cell RNA-seq characterisation of the haemogenic gastruloid at 120h and have not looked specifically at earlier timepoints of embryo patterning. This said, we show the presence of neuroectodermal cells in cluster 9; on the other hand, cluster 7 includes hepatoblast-like cells, denoting endodermal specification (Supplementary File S2). However, in the absence of earlier timepoints and given the bias towards mesodermal specification, we expect that specification of ectodermal and endodermal programmes may be incomplete.

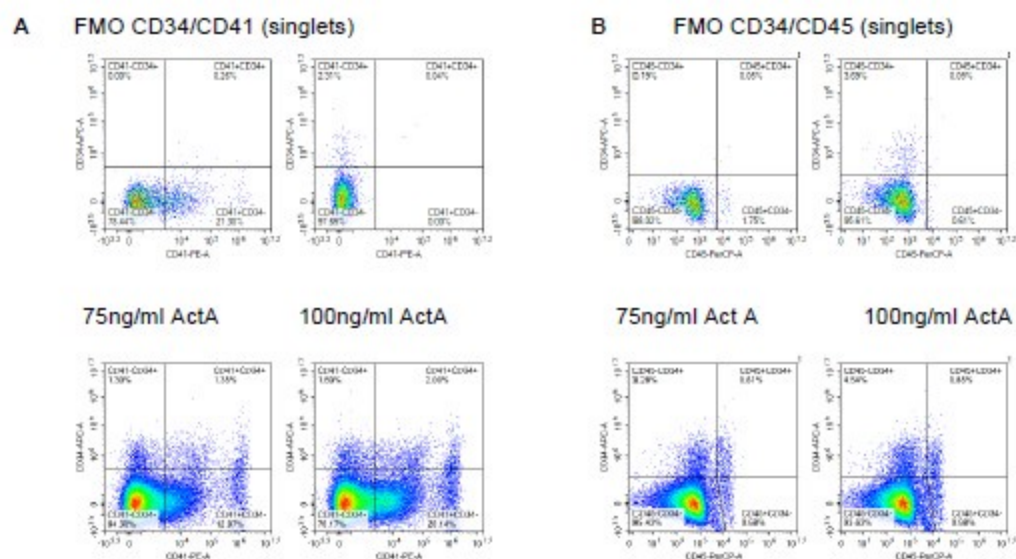
In respect of the contention regarding the capture of YS-like and AGM-like haematopoiesis, we had presented evidence in the original version of the manuscript that haemogenic cells generated during gastruloid differentiation, particularly at late 192h and 216h timepoints project onto highly purified CKit+ CD31+ Gfi1-expressing cells from mouse AGM (PMID: 38383534), providing support for at least partial recapitulation of the corresponding developmental stage. These projections are represented in Fig. 4A, right and 4S1C of the revised manuscript. In distinguishing between YS-like and AGM-like haematopoiesis, we call

the Reviewer's attention to the replotting of the single-cell RNA-seq data already in the manuscript, which we provided in response to point 1 (Fig. 3B-D and 3S2B), which highlights an increase in Sox17, but not Sox18, expression in the 192h haemogenic endothelium, which suggests an association with AGM haematopoiesis (PMID: 20228271). A significant association of Cd44 and Procr expression with the same time-point (Fig. 3B-D in the manuscript), further supports an AGM-like endothelial-to-haematopoietic transition at the 192h timepoint. We have re-analysed the scRNA-seq data to better represent the expression of these markers in Fig. 3A-E and S32B. We agree that it remains challenging to identify markers exclusive to AGM haematopoiesis, which is operationally equated with generation of transplantable haematopoietic stem cells. While HSC generation is a key event characteristic of the AGM, not all AGM haematopoiesis corresponds to HSCs, an important point in evaluating the data presented in the manuscript, and one that is acknowledged by us. The main text has been edited to clarify the experiments pertaining to distinguishing AGM and YS haematopoiesis, which are detailed in lines 180-187, 200-221, 268-304, and 315-356.

Following on the Reviewer's comments about Cd34, we also inspected co-expression of Cd34 with Cd41 and Cd45, the latter co-expression present in, although not necessarily exclusive to, AGM haematopoiesis. Reassuringly, we observed clear co-expression with both markers (Author response image 1), in addition to a CD41+CD34- population, which likely reflects YS EMP-independent erythropoiesis. Flow cytometry analysis of co-expression of CD31 and CD34 in CD41+ and CD45+ populations at 144h and 216h timepoints has been included in Fig. 2B-D, Fig. 2S1A-D, including controls. In text: 180-187. We have earlier on in the rebuttal highlighted the fact that marker expression is responsive to the levels of Activin A used in the patterning pulse, with the 100ng/ml Activin A used in our protocol superior to 75ng/ml.

Author response image 1.

Association of CD34 with CD41 and CD45 expression is Activin A-responsive and supports the presence of definitive haematopoiesis. A. Flow cytometry analysis of CD34 and CD41 expression in 216h-haemogenic gastruloids; two doses of Activin A were used in the patterning pulse with CHI99021 between 48-72h. FMO controls shown. B. Flow cytometry analysis of CD34 and CD45 at 216h in the same experimental conditions.



Given the centrality of this point in comments by all the Reviewers, we have conducted projections of our single-cell RNA-seq data against two studies which (1) capture arterial and haemogenic specification in the para-splanchnopleura (pSP) and AGM region between E8.0

and E11 (Hou et al, PMID: 32203131), and (2) uniquely capture YS, AGM and FL progenitors and the AGM endothelial-to-haematopoietic transition (EHT) in the same scRNA-seq dataset (Zhu et al, PMID: 32392346). Focusing the analysis on the subsets of haemogenic gastruloid cells sorted as CD41+ (144h) C-Kit+ (144h and 192h) and CD45+ (192h and 216h) (now represented in Fig. 3A, and projected onto the studies in Fig. 4A), we show:

(1) That a subset of haemato-endothelial cells from haemogenic gastruloids at 144h to 216h project onto intra-embryonic cells spanning E8.25 to E10 (revised Fig. 4A left and 4S1A). This is in agreement with our original interpretation that 216h are no later than the MPP/pre-HSC state of embryonic development, requiring further maturation to generate engrafting progenitors. We have nevertheless removed specific references to pre-HSC, and instead referred to HSPC/progenitors.

(2) That haemogenic gastruloids contain YS-like (including EMP-like) and AGM-like haematopoietic cells (Fig. 4A centre and 4S1B). Significantly, some of the cells, particularly CKit-sorted cells with a candidate endothelial and HE-like signature project onto AGM pre-HE and HE, as well as IAHC. Some 144h CD41+ and 192h CD45+ cells also project onto IAHC, suggesting that YS-like and AGM-like programmes arise independently and with partial timedependent organisation in the haemogenic gastruloid model. Later, predominantly 216h cells, have characteristics of MPP/LMPP-like cells from the FL, suggesting a progenitor wave of differentiation.

Altogether, the data support the notion that haemogenic gastruloids capture YS and AGM haematopoiesis until E10, as suggested by us in the manuscript. This re-analysis of the scRNA-seq data which was indeed prompted by challenging and insightful comments from the Reviewers, has been incorporated in the manuscript as described above and further listed here:

Re-clustering and highlights of specific markers in our scRNA-seq data in Figure 3A-E. In text: 268-304.

Projections to mouse embryo datasets in Figure 4A (Figure 4S1A-C; Supplementary File 3). In text: 315-356.

Single-cell RNA sequencing was used to compare hGx with mouse AGM. The authors incorrectly conclude that '...specification of endothelial and HE cells in hGx follows with time-dependent developmental progression into putative AGM-like HE.' And, '...HE-projected hGx cells.....expressed Gata2 but not Runx1, Myb, or Gfi1b..' Hemogenic endothelium is defined by the expression of Runx1 and Gfi1b is downstream of Runx1.

As a hierarchy of regulation, Gata2 precedes and drives Runx1 expression at the specification of HE (PMID: 17823307; PMID: 24297996), while Runx1 drives the EHT, upstream of Gfi1b in haematopoietic clusters (PMID: 34517413). Please note that the text segment the Reviewer refers to has been removed from the manuscript, as the analysis is no longer solely focused on projection to Thambyrajah et al (2024) data, and instead gained significantly from the projections on to the Hou et al (2020) and Zhu et al (2020) studies, as detailed above.

(3) The hGx protocol 'generates hematopoietic SC precursors capable of short-term engraftment' is not supported by the data presented. Short-term engraftment would be confirmed by flow cytometric detection of hematopoietic cells within the recipient bone marrow, spleen, thymus, and peripheral blood that expressed the BFP transgene. This analysis was not provided. PCR detection of transcripts, following an unspecified number of amplification cycles, as shown in Figure 3G (incorrectly referred to as Figure 3F in the legend) is not acceptable evidence for engraftment.

We provide the full flow cytometry analysis of spleen engraftment in the 5 mice which received implantation of 216h-haemogenic gastruloids in the adrenal gland and were analysed at 4 weeks; an additional (control) animal received adrenal injection of PBS (Fig. 4B-D in the revised manuscript). In this experiment, the bone marrow collection was limiting, and material was prioritised for PCR (Fig. 4C and full gels in 4S2C in the revised manuscript).

We had previously provided only representative plots of flow cytometry analysis of bone marrow and spleen, which we described as low-level engraftment and were chosen conservatively. The analysis was meant to complement the genomic DNA PCR, where detection was present in only some of the replicates tested per animal. On this note, we confirm that PCR analysis used conventional 40 cycles; the sensitivity had already been shown in the earlier version of the manuscript and is again represented in Fig. 4S2B. We argue that the low level of cytometric and molecular engraftment at 4 weeks, from haemogenic gastruloid-derived progenitors that have not progressed beyond a stage equivalent to E10 (Fig. 4A and Supplementary File 3 in the revised manuscript from scRNAseq projections), and that we have described as requiring additional maturation in vivo, are not surprising. Indeed, as previously shown and now repeated in Fig. 2B-E (controls in Fig. 2S1E-G) in the revised manuscript, no more than 7 CD45⁺CD144⁺ multipotent cells are present per haemogenic gastruloid. We are only able to implant 3 haemogenic gastruloids in the adrenal gland of each transplanted animal.

We have rephrased Results and Discussion in lines 359-415 and 588-621, respectively, to rectify the nature of the engraftment, which we now attribute more generically to progenitors, also in light of the developmental time we could capture in the gastruloids prior to implantation.

Transplanted hGx formed teratoma-like structures, with hematopoietic cells present at the site of transplant only analysed histologically. Indeed, the quality of the images provided does not provide convincing validation that donor-derived hematopoietic cells were present in the grafts.

As stated in the text, the images mean to illustrate that the haemogenic gastruloids developed in situ. Further analysis motivated by the Reviewers' comments and indeed a subsequent experiment with analysis of engraftment at a later timepoint of 8 weeks (revised Fig. 4E and 4S2F-G) did not show a direct correspondence between engraftment and in vivo development or expansion, although this occurs in some cases. To be clearer, the observation of donor-derived blood cells in the implanted haemogenic gastruloids would not correspond to engraftment, as we have amply demonstrated that they have generated blood cells in vitro. There is no evidence that there are remaining pluripotent cells in the haemogenic gastruloid after 9 days of differentiation, and it is therefore not clear that the structures observed are teratomas. We specifically comment on this point in the revised manuscript – lines 601-607.

There is no justification for the authors' conclusion that '... the data suggest that 216h hGx generate AGM-like pre-HSC capable of at least short-term multilineage engraftment upon maturation...'. Indeed, this statement is in conflict with previous studies demonstrating that pre-HSCs in the dorsal aorta of the mouse embryo are immature and actually incapable of engraftment.

We have clearly stated that we do not see haematopoietic engraftment through transplantation of dissociated haemogenic gastruloids, which reach the E10 state containing pre-HSC (revised Fig 4A, 4S1A and Supplementary File 3). Instead, we observed rare myelo-erythroid (revised Fig. 4S2F-G) and myelo-lymphoid (revised Fig. 4E) engraftment upon in vivo maturation of haemogenic gastruloids with preserved 3D organisation. These statements are not contradictory. Nevertheless, we have now more cautiously attributed engraftment to

the present of progenitors as a generic designation, and not to pre-HSC (lines 412-414 and 588-592 in the revised manuscript).

The statement '...low-level production of engrafting cells recapitulates their rarity in vivo, in agreement with the embryo-like qualities of the gastruloid system....' is incorrect. Firstly, no evidence has been provided to show the hGx has formed a dorsal aorta facsimile capable of generating cells with engrafting capacity. Secondly, although engrafting cells are rare in the AGM, approximately one per embryo, they are capable of robust and extensive engraftment upon transplantation.

As indicated above, the statement in lines 412-414 now reads “Engraftment is erythromyeloid at 4 weeks and lympho-myeloid at 8 weeks, reflecting different classes of progenitors, putatively of YS-like and AGM-like affiliation.” To be clear, with our original statement we meant to highlight that the production of definitive AGM-like haematopoietic progenitors (not all of which are engrafting) in haemogenic gastruloids does not correspond to non-physiological single-lineage programming. We did and do not claim that we achieved production of HSC, which would be long-term engrafting.

(4) Expression MNX1 transcript and protein in hematopoietic cells in MNX1 rearranged acute myeloid leukaemia (AML) is one cause of AML in infants. In the hGX model of this disease, Mnx1 is overexpressed in the mESCs that are used to form gastruloids. Mnx1 overexpression seems to confer an overall growth advantage on the hGx and increase the serial replating capacity of the small number of hematopoietic cells that are generated. The inefficiency with which the hGx model generates hematopoietic cells makes it difficult to model this disease. The poor quality of the cytopsin images prevents accurate identification of cells. The statement that the kit-expressing cells represent leukemic blast cells is not sufficiently validated to support this conclusion. What other stem cell genes are expressed? Surface kit expression also marks mast cells, frequently seen in clonogenic assays of blood cells. Flow cytometric and gene expression analyses using known markers would be required.

The haemogenic gastruloid model generates haematopoietic and haemato-endothelial cells. MNX1 expands C-Kit⁺ cells at 144h, which we show to have a haemato-endothelial signature (see revised Fig. 3A-E, Supplementary File 2). We have added additional flow cytometry data showing that the replating cells from MNX1 express CD31 (Figure 6S1A-B).

Serial replating of CFC assays is a conventional in vitro assay of leukaemia transformation. Critically, colony replating is not maintained in EV control cells, attesting to the transformation potential of MNX1. Although we have not fully-traced the cellular hierarchy of MNX1-driven transformation in the haemogenic gastruloid system, the in vitro replating expands a C-Kit⁺ cell (revised Fig. 6E), which reflects the surface phenotype of the leukaemia, also recapitulated in the mouse model initiated by MNX1-overexpressing FL cells. Importantly, it recapitulates the transcriptional profile of MNX1leukaemia patients (revised Fig. 7C), which is uniquely expressed by MNX1144h and replated colony cells, but not to MNX1 216h gastruloid cells, arguing against a generic signature of MNX1 overexpression (revised Fig. 7B). Importantly, the MNX1-transformation of haemogenic gastruloid cells is superior to the FL leukaemia model at capturing the unique transcriptional features of MNX1-driven leukaemia, distinct from other forms of AML in the same age group (Fig 7 S1D-F). It is possible that this corresponds to a pre-leukaemia event, and we will explore this in future studies, which are beyond the proof-of-principle nature of this paper.

(5) In human infant MNX1 AML, the mutation is thought to arise at the fetal liver stage of development. There is no evidence that this developmental stage is mimicked in the hGx model.

We never claim that the haemogenic gastruloid model mimics the foetal liver. We propose that susceptibility to MNX1 is at the HE-to-EMP transition. Moreover, and importantly, contrary to the Reviewer's statement, there is no evidence in the literature that the mutation arises in the foetal liver stage, just that the mutation arises before birth (PMID: 38806630), which is different. In a mouse model of MNX1 overexpression, the authors achieve leukaemia engraftment upon MNX1 overexpression in foetal liver, but not in bone marrow cells (PMID: 37317878). This is in agreement with a vulnerability of embryonic / foetal, but not adult cells to the MNX1 expression caused by the translocation. However, haematopoietic cells in the foetal liver originate from YS and AGM precursors, so the origin of the MNX1susceptible cells can be in those locations, rather than the foetal liver itself.

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors develop an exciting new hemogenic gastruloid (hGX) system, which they claim reproduces the sequential generation of various blood cell types. The key advantage of this cellular system would be its potential to more accurately recapitulate the spatiotemporal emergence of hematopoietic progenitors within their physiological niche compared to other available in vitro systems. The authors present a large set of data and also validate their new system in the context of investigating infant leukemia.

Strengths:

The development of this new in vitro system for generating hematopoietic cells is innovative and addresses a significant drawback of current in vitro models. The authors present a substantial dataset to characterize this system, and they also validate its application in the context of investigating infant leukemia.

Weaknesses:

The thorough characterization and full demonstration that the cells produced truly represent distinct waves of hematopoietic progenitors are incomplete. The data presented to support the generation of late yolk sac (YS) progenitors, such as lymphoid cells, and aortic-gonad-mesonephros (AGM)-like progenitors, including pre-hematopoietic stem cells (pre-HSCs), by this system are not entirely convincing. Given that this is likely the manuscript's most crucial claim, it warrants further scrutiny and direct experimental validation. Ideally, the identity of these progenitors should be further demonstrated by directly assessing their ability to differentiate into lymphoid cells or fully functional HSCs. Instead, the authors primarily rely on scRNA-seq data and a very limited set of markers (e.g., Ikzf1 and Mllt3) to infer the identity and functionality of these cells. Many of these markers are shared among various types of blood progenitors, and only a well-defined combination of markers could offer some assurance of the lymphoid and pre-HSC nature of these cells, although this would still be limited in the absence of functional assays.

The identification of a pre-HSC-like CD45⁺CD41⁻/lo C-Kit⁺VE-Cadherin⁺ cell population is presented as evidence supporting the generation of pre-HSCs by this system, but this claim is questionable. This FACS profile may also be present in progenitors generated in the yolk sac such as early erythromyeloid progenitors (EMPs). It is only within the AGM context, and in conjunction with further functional assays demonstrating the ability of these cells to differentiate into HSCs and contribute to long-term repopulation, that this profile could be strongly associated with pre-HSCs. In the absence of such data, the cells exhibiting this profile in the current system cannot be conclusively identified as true pre-HSCs.

We present 2 additional pieces of evidence to support our claims that we capture YS and AGM stages of haematopoietic development.

(I) In the new Figures 4A and 4 S1A-C and Supplementary File 3 in the revised manuscript, we project our single-cell RNA-seq data onto (1) developing intra-embryonic pSP and AGM between E8 and E11 (Fig. 4A left, 4S1A) and (2) a single-cell RNA-seq study of HE development which combines haemogenic and haematopoietic cells from the YS, the developing HE and IAHC in the AGM, and FL (Fig. 4A centre, 4S1B). Our data maps E8.25-E10, and captures YS EMP and erythroid and myeloid progenitors, as well as AGM pre-HE, HE and IAHC, with some cells matching HSPC and LMPP, as suggested by the projection onto the Thambyrajah et al data set (already presented in the previous version of the manuscript, and now in Fig. 4A right and 4 S1C). The projection of the scRNA-seq data is presented in lines 314-355 of the revised manuscript. The scRNA-seq data itself was refocused on haemato-endothelial programmes as presented in the revised Fig. 3A-E, described in lines 267-303.

(II) Given the difficulty in finding markers that specifically associate with AGM haematopoiesis, we inspected the possibility of capturing different regulatory requirements at different stages of gastruloid development mirroring differential effects in the embryo. Polycomb EZH2 is specifically required for EMP differentiation in the YS, but does not affect AGM-derived haematopoiesis; it is also not required for primitive erythroid cells (PMID: 29555646; PMID: 34857757). We treated haemogenic gastruloids from 120h onwards with either DMSO (0.05%) or GSK126 (0.5uM), and inspected the cellularity of gastruloids at 144h, which we equate with YS-EMP, and 216h – putatively AGM haematopoiesis. We show that EZH2 inhibition / GSK126 treatment specifically reduces %CD41+ cells at 144h, but does not reduce %CD41+ or %CD45+ cells at 216h. We have included this experiment in the manuscript in Fig. 2 S2B-C (in text: 209-221).

These data, together with the scRNA-seq projections described, provide evidence to our claim that 144h haemogenic gastruloids capture YS EMPs, while CD41+ and CD45+ cells isolated at 216h reflect AGM progenitors. We cannot conclude as to the functional nature of the AGM cells from this experiment. The main text has been edited to clarify the experiments pertaining to distinguishing AGM and YS haematopoiesis (lines 180-187; 200-221; 268-304; 315-356).

The engraftment data presented are also not fully convincing, as the observed repopulation is very limited and evaluated only at 4 weeks post-transplantation. The cells detected after 4 weeks could represent the progeny of EMPs that have been shown to provide transient repopulation rather than true HSCs.

In the original version of the manuscript, we stated that there is low level engraftment and did not claim to have generated HSC. Instead, we described cells with short-term engraftment potential. We agree with the Reviewer that the cells we show in the manuscript at 4 weeks could be EMPs (revised Fig. 4B-E and 4 S2D-G). Additionally, we now have 8-week analysis of implant recipients, in which we observed, again low-level, a multi-lineage engraftment of the recipient bone marrow in 1:3 recipients (revised Fig. 4B-E and 4S2F-H). This engraftment is myeloid-lymphoid and therefore likely to have originated in a later progenitor. To be clear, we do not claim that this corresponds to the presence of HSC. It nevertheless supports the maturation of progenitors with engraftment potential. Limiting amounts of material was prioritised for flow cytometry stainings, not allowing PCR analysis. We rephrased Results and Discussion in lines 359-414 and 588-621, respectively, to rectify the nature of the engraftment.

Reviewer #3 (Public review):

In this study, the authors employ a mouse ES-derived "hemogenic gastruloid" model which they generated and which they claim to be able to deconvolute YS and AGM stages

of blood production *in vitro*. This work could represent a valuable resource for the field. However, in general, I find the conclusions in this manuscript poorly supported by the data presented. Importantly, it isn't clear what exactly are the "YS" and the "AGM"-like stages identified in the culture and where is the data that backs up this claim. In my opinion, the data in this manuscript lack convincing evidence that can enable us to identify what kind of hematopoietic progenitor cells are generated in this system. Therefore, the statement that "our study has positioned the MNX1-OE target cell within the YS-EMP stage (line 540)" is not supported by the evidence presented in this study. Overall, the system seems to be very preliminary and requires further optimization before those claims can be made.

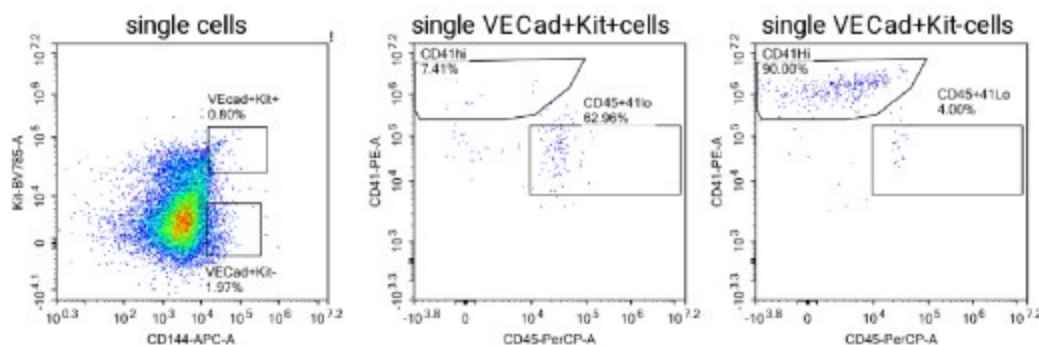
Specific comments below:

(1) The flow cytometric analysis of gastruloids presented in Figure 1 C-D is puzzling. There is a large % of C-Kit+ cells generated, but few VE-Cad+ Kit+ double positive cells. Similarly, there are many CD41+ cells, but very few CD45+ cells, which one would expect to appear toward the end of the differentiation process if blood cells are actually generated. It would be useful to present this analysis as consecutive gating (i.e. evaluating CD41 and CD45 within VE-Cad+ Kit+ cells, especially if the authors think that the presence of VE-Cad+ Kit+ cells is suggestive of EHT). The quantification presented in D is misleading as the scale of each graph is different.

Fig. 1C-D provide an overview of haemogenic markers during the timecourse of haemogenic gastruloid differentiation, and does indeed show a late up-regulation of CD45, as the Reviewer points out would be expected. The %CD45+ cells is indeed low. However, we should point out that the haemogenic gastruloid protocol, although biased towards mesodermal outputs, does not aim to achieve pure haematopoietic specification, but rather place it in its embryo-like context. We refute that the scale is misleading: it is a necessity to represent the data in a way that is interpretable by the reader: and we made sure from the outset that the gates (in C) are truly representative and annotated, as are the plot axes (in D). Consecutive gating at the 216h-timepoint is shown and quantified in Fig. 2S1D-F, or in the alternative consecutive gating suggested by the Reviewer, in Author response image 2 below. At the request of Reviewer 1, we also analysed CD31 and CD34 within CD41 and CD45 populations, again as validation of the emergent haematopoietic character of the cells obtained. This new analysis is shown in revised Fig. 2B, quantified in 2C.

Author response image 2.

Flow cytometry analysis of VE-cadherin+ cells in haemogenic gastruloids at 216h of the differentiation protocol, probing co-expression of CD45, CD41 and C-Kit.

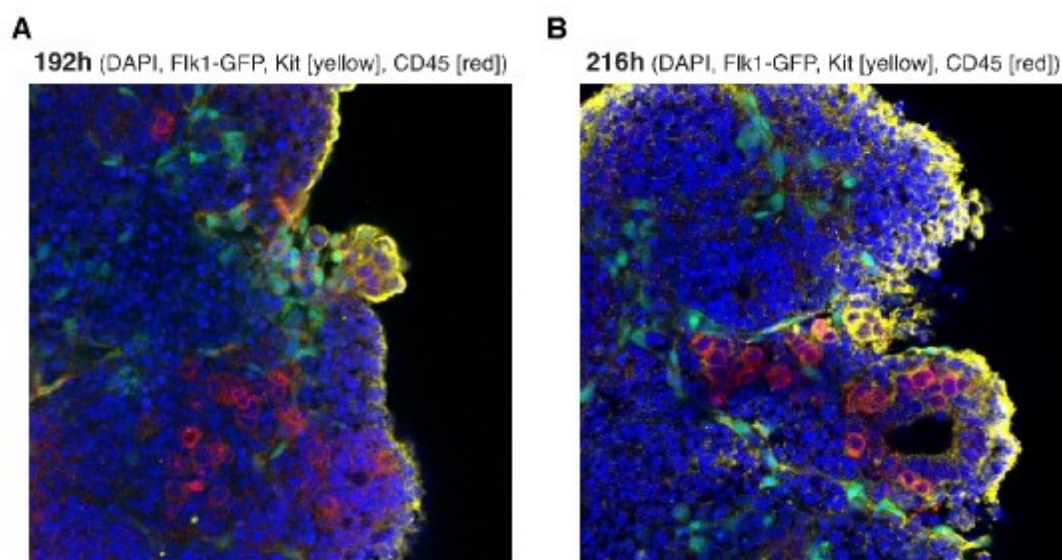


(2) The imaging presented in Figure 1E is very unconvincing. C-Kit and CD45 signals appear as speckles and not as membrane/cell surfaces as they should. This experiment should be repeated and nuclear stain (i.e. DAPI) should be included.

We included the requested immunofluorescence staining in Figure 1E (216h). We also show the earlier timepoint of 192h here as Author response image 3. In text: lines 158-162.

Author response image 3.

Confocal images of haematopoietic production in haemogenic gastruloids. Wholemout, cleared haemogenic gastruloids were stained for CD45 (pseudo-coloured red) and C-Kit antigens (pseudo-coloured yellow) with indirect staining, as described in the manuscript. Flk1-GFP signal is shown in green. Nuclei are contrasted with DAPI. (A) 192h. (B) 216h.



(3) Overall, I am not convinced that hematopoietic cells are consistently generated in these organoids. The authors should sort hematopoietic cells and perform May-Grunwald Giemsa stainings as they did in Figure 6 to confirm the nature of the blood cells generated.

It is factual that the data are reproducible and complemented by functional assays shown in revised Fig. 2D-E, which clearly demonstrate haematopoietic output. The single-cell RNA-seq data also show expression of a haematopoietic programme, which we have complemented with biologically independent qRT-PCR analysis of the expression of key endothelial and haematopoietic marker and regulatory genes (revised Fig. 2F; in text: 200-209). As requested, we include Giemsa-Wright's stained cytopspins obtained at 216h to illustrate haematopoietic output. These are shown in revised Fig. 2S2A, in text: lines 194-199. Inevitably, the cytopspins will be inconclusive as to the presence of endothelial-tohaematopoietic transition or the generation of haematopoietic stem/progenitor cells, as these cells do not have a distinctive morphology.

(4) The scRNAseq in Figure 2 is very difficult to interpret. Specific points related to this: - Cluster annotation in Figure 2a is missing and should be included.

Why do the heatmaps show the expression of genes within sorted cells? Couldn't the authors show expression within clusters of hematopoietic cells as identified transcriptionally (which ones are they? See previous point)? Gene names are illegible.

I see no expression of Hlf or Myb in CD45+ cells (Figure 2G). Hlf is not expressed by any of the populations examined (panels E, F, G). This suggests no MPP or pre-HSC are generated in the culture, contrary to what is stated in lines 242-245. (PMID 31076455 and 34589491). Later on, it is again stated that "hGx cells... lacked detection of HSC genes like Hlf, Gfi1, or Hoxa9" (lines 281-283). To me, this is proof of the absence of AGM-like hematopoiesis generated in those gastruloids.

For a combination of logistic and technical reasons, we performed single-cell RNA-seq using the Smart-Seq2 platform, which is inherently low throughput. We overcame the issue of cell coverage by complementing whole-gastruloid transcriptional profiling at successive time-points with sorting of subpopulations of cells based on individual markers documented in Fig. 1. We clearly stated which platform was used as well as the number and type of cells profiled (Fig. 3S1 and lines 226-241 of the revised manuscript), and our approach is standard. Following suggestions of the Reviewers to further focus our analysis on the haemogenic cellular differentiation within the gastruloids, we revised the presentation of the scRNA-seq data to now provide UMAP projections with representation and quantification of individual genes, including the ones queried by the Reviewer in Fig. 3 and respective supplements. Specifically, re-clustering and highlighting of specific markers are shown in Figure 3A-D and presented in lines 267-303 of the revised manuscript. Complementary independent real-time quantitative (q)PCR analysis showing time-dependent expression of endothelial and haematopoietic markers is now in Figure 2F. In text: 200-208.

(5) Mapping of scRNA-Seq data onto the dataset by Thambyrajah et al. is not proof of the generation of AGM HE. The dataset they are mapping to only contains AGM cells, therefore cells do not have the option to map onto something that is not AGM. The authors should try mapping to other publicly available datasets also including YS cells.

We have done this and the data are presented in Figure 4A (Figure 4S1A) and Supplementary File. In text: 314-355. As detailed in response to Reviewer 1, we have conducted projections of our single-cell RNA-seq data against two studies which (1) capture arterial and haemogenic specification in the para-splanchnopleura (pSP) and AGM region between E8.0 and E11 (Hou et al, PMID: 32203131) (revised Fig. 4A and 4 S1A), and (2) uniquely capture YS, AGM and FL progenitors and the AGM endothelial-to-haematopoietic transition (EHT) in the same scRNA-seq dataset (Zhu et al, PMID: 32392346) (revised Fig. 4A and 4 S1B). Specifically in answering the Reviewers' point, we show that different subsets of haemogenic gastruloid cells sorted on haemogenic surface markers C-Kit, CD41 and CD45 cluster onto pre-HE and HE, intra-aortic clusters and FL progenitor compartments, and to YS EMP and erythroid and myeloid progenitors. This lends support to our claim that the haemogenic gastruloid system specifies both YS-like and AGM-like cells. Please note that we now do point out that some CD41+ cells at 144h project onto IAC, as do cells at the later timepoints, suggesting that AGM-like and YS-EMP-like waves may overlap at the 144h timepoint (lines...). In the future, we will address specific location of these cells, but that corresponds to a largescale spatial transcriptomics analysis requiring extensive optimisation for section capture which is beyond the scope of this manuscript and this revision.

(6) Conclusions in Figure 3, named "hGx specify cells with preHSC characteristics" are not supported by the data presented here. Again, I am not convinced that hematopoietic cells can be efficiently generated in this system, and certainly not HSCs or pre-HSCs.

We have provided evidence in the original manuscript, and now through additional experiments, that there is haematopoietic specification, including of progenitor cells, in the haemogenic gastruloid system. Molecular markers are shown in revised Fig. 2F and Fig. 3 and supplements; CFC assays are shown in revised Fig. 2D-E; cytopins are in revised Fig. 2 S2A; further analysis of 4-week implants and new analysis of 8-week implants (discussed below) are in revised Fig. 4 B-D and Fig. 4 S2 and we discussed the new scRNA-seq projections above. Importantly, we have never claimed, and again do not, that haemogenic gastruloids generate HSC. We accept the Reviewer's comment that we have not provided sufficient evidence for the specification of pre-HSC-like cells and accordingly now refer more generically and conservatively to progenitors.

FACS analysis in 3A is again very unconvincing. I do not think the population identified as C-Kit⁺ CD144⁺ is real. Also, why not try gating the other way around, as commonly done (e.g. VE-Cad⁺ Kit⁺ and then CD41/CD45)?

Our gating strategy is not unconventional, which was done from a more populated gate onto the less abundant one to ensure that the results are numerically more robust. In the case of haemogenic gastruloids, unlike the AGM preparations the Reviewer may be referring to, CD41 and CD45⁺ cells are more abundant as there is no circulation of more differentiated haematopoietic cells away from the endothelial structures. This said, we did perform the gating as suggested (Rev Fig. 2), indeed confirming that most VE-cad⁺ Kit⁺ cells are CD45⁺. Interestingly VE-cad⁺Kit⁻ are predominantly CD41⁺, reinforcing the haematopoietic nature of these cells.

The authors must have tried really hard, but the lack of short- or long-engraftment in a number of immunodeficient mouse models (lines 305-313) really suggests that no blood progenitors are generated in their system. I am not familiar with the adrenal gland transplant system, but it seems like a very non-physiological system for trying to assess the maturation of putative pre-HSCs. The data supporting the engraftment of these mice, essentially seen only by PCR and in some cases with a very low threshold for detection, are very weak, and again unconvincing. It is stated that "BFP engraftment of the Spl and BM by flow cytometry was very low level albeit consistently above control (Fig. S4E)" (lines 337-338). I do not think that two dots in a dot plot can be presented as evidence of engraftment.

We have presented the data with full disclosure and do not deny that the engraftment achieved is low-level and short-term, indicating incomplete maturation of definitive haematopoietic progenitors in the current haemogenic gastruloid system. Indeed, by not wanting to overstate the finding, we were deliberately conservative in our representative flow cytometry plots and focused on the PCR for sensitivity. We now present the full flow cytometry analysis for spleen where we preserved more cells after the genomic DNA extraction (revised Fig. 4C) and call the Reviewer's attention to the fact that detection of BFP⁺ cells by PCR and flow cytometry in the recipient animals is consistent between the 2 methods (revised Fig. 4C and D; full gels previously presented now in Fig. 4S2C; sensitivity analysis was also previously available and is now in Fig. 4S2B). In addition, we have now also been able to detect low-level myelo-lymphoid engraftment in the bone marrow and spleen 8 weeks after adrenal implantation, again suggesting the presence of a small number of definitive haematopoietic progenitors that potentially mature from the 3 haemogenic gastruloids implanted (Fig. 4E and 4 S2F-G in the revised manuscript. We rephrased Results and Discussion at lines 359-414 and 589-621, respectively, to rectify the nature of the engraftment which we attribute to progenitors.

(7) Given the above, I find that the foundations needed for extracting meaningful data from the system when perturbed are very shaky at best. Nevertheless, the authors proceed to overexpress MNX1 by LV transduction, a system previously shown to transform fetal liver cells, mimicking the effect of the t(7;12) AML-associated translocation. Comments on this section:

The increase in the size of the organoid when MNX1 is expressed is a very unspecific finding and not necessarily an indication of any hematopoietic effect of MNX1 OE.

We agree with the Reviewer on this point; it is nevertheless a reproducible observation which we thought relevant to describe for completeness and data reproducibility.

The mild increase of cKit+ cells (Figure 4E) at the 144hr timepoint and the lack of any changes in CD41+ or CD45+ cells suggests that the increase in Kit+ cells % is not due to any hematopoietic effect of MNX1 OE. No hematopoietic GO categories are seen in RNA seq analysis, which supports this interpretation. Could it be that just endothelial cells are being generated?

The Reviewer is correct that the MNX1-overexpressing cells have a strong endothelial signature, which is present in patients (revised Fig. 5A). We investigated a potential link with C-Kit by staining cells from the replating colonies during the process of in vitro transformation with CD31. We observed that 40-50% of C-Kit+ cells (20-30% total colony cells) co-expressed CD31, at least at early plating. These cells co-exist with haematopoietic cells, namely Ter119+ cells, as expected from the YSlike erythroid and EMP-like affiliation of haematopoietic output from 144h-haemogenic gastruloids. These data are included in Fig. 6S1A-B (in text 506-507) of the revised manuscript.

(8) There seems to be a relatively convincing increase in replating potential upon MNX1-OE, but this experiment has been poorly characterized. What type of colonies are generated? What exactly is the "proportion of colony forming cells" in Figures 5B-D? The colony increase is accompanied by an increase in Kit+ cells; however, the flow cytometry analysis has not been quantified.

Given the inability to replate control EV cells, there is not a population to compare with in terms of quantification. The level of C-Kit+ represented in Fig. 6E of the revised manuscript is achieved at plate 2 or 3 (depending on the experiment), both of which are significantly enriched for colony-forming cells relative to control (revised Fig. 6B, D).

(9) Do hGx cells engraft upon MNX1-OE? This experiment, which appears not to have been performed, is essential to conclude that leukemic transformation has occurred.

For the purpose of this study, we are satisfied with confirmation of in vitro transformation potential of MNX1 haemogenic gastruloids, which can be used for screening purposes. Although interesting, in vivo leukaemia engraftment from haemogenic gastruloids is beyond the scope of this study.

Reviewer #2 (Recommendations for the authors):

(1) Minor comments

(a) I find the denomination "hGx" very confusing as it would suggest that these gastruloids are human, whereas, in fact, they are murine.

We agree with the Reviewer on the confusing nomenclature and have edited the manuscript to call “haemGx” instead.

(b) I find the presence of mast cells in CFC of MNX1-OE cultures very puzzling as this does not bear any resemblance to human leukemia.

We detect an enrichment of mast cell transcriptional programmes, as defined by the cell type repositories. While it is not mast cells to represent leukaemic cells in patients, this ontology is likely to reflect the developmental stage and origin of progenitors which are affected by MNX1.

(2) I have a few suggestions to improve figures and tables clarity, to help readers better follow the data presented.

(a) To enhance readability, it would be beneficial to highlight the genes mentioned in the text within the scRNA-seq figures. Many figures currently display over 30-40 genes in small font sizes, making it difficult to quickly locate specific genes discussed in the text. Additionally, implementing a colorcoding system to categorize these genes according to their proposed lineages would improve clarity and organization.

We have now performed major re-organisation and re-analyses of the scRNA-seq data, which we believe has improved the readability and clarity of the corresponding sections of the manuscript.

(b) The data presented in Supplementary Table 1, along with other supplementary tables, are challenging to interpret due to insufficient annotations. Enhancing these tables with clearer and more detailed annotations would significantly improve clarity and aid readers in understanding the supplementary materials.

Descriptive text has been added to accompany each Supplementary File to aid in understanding the results reported therein.

Reviewer #3 (Recommendations for the authors):

In addition to what was written in the public review, I would suggest the authors simplify and shorten the text. Currently, a lot of unnecessary detail is included which makes the story very hard to follow. Moreover, the authors should modify the figures to make them more comprehensible, especially for RNA-seq data.

We have significantly re-arranged and shortened parts of the manuscript, particularly by focusing the Discussion. Results presentation has also been improved through additional analysis and graphic representation of the scRNA-seq data, which we believe has improved the readability and clarity.

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